PlotExtractor

User guide

Make and edit plot grids, cut one original-quality image per plot from Phase One drone photos (from the JPEGs or straight from the IIQ raw files), and make training tiles from the middle crop rows of each plot.

Department of Primary Industries and Regional Development
GOVERNMENT OF WESTERN AUSTRALIA

APPN
Australian Plant Phenomics Network

PlotExtractor 2026-10-24

One original-quality image per plot, and training tiles from the middle rows

Phase One RGB · Metashape project + plot shapefile · DPIRD / APPN

raw plot images (original pixels)

middle-3-row tiles, plot 3003

PlotExtractor 2026-10-30 · DPIRD / APPN · 30 October 2026

Contents

1. What PlotExtractor does

2. Quick start: the recommended way

3. Installing and starting

4. Tab 1 · Generate plot grid

5. Tab 2 · Edit plot grid

6. Tab 3 · IIQ conversion

7. Tab 4 · Extract plots

- 7.1 The fields
- 7.2 Raw crop styles: which one to use
- 7.3 What you get
- 7.4 Check the coverage

8. Tab 5 · IIQ plot extraction

- 8.1 Bits and compression
- 8.2 How a plot is made from the IIQ, step by step

9. Tab 6 · Training tiles

- 9.1 Tab 6a · Plot tiles of a fixed ground size
- 9.2 Tab 6b · Whole rows and row tiles
- 9.3 The rows numbering
- 9.4 The recipe file
- 9.5 How the rows are found
- 9.6 Where the tiles go along the plot
- 9.7 What you get (tab 6a)
- 9.8 Other tools on this tab

10. Tab 7 · QA, and are the images really original?

11. Tab 8 · Acknowledgements

12. Tips and good practice

- 12.1 Test first
- 12.2 Use a small margin
- 12.3 Give the sown rows
- 12.4 Flight planning for whole plots in one photo
- 12.5 Check the QA pictures
- 12.6 Use one output scale across trials

13. Troubleshooting

Appendix: files

1. What PlotExtractor does

PlotExtractor takes a drone flight that has been processed in Agisoft Metashape and gives you a plot grid, one image per plot and training tiles. The tabs are numbered in the order you normally use them:

Tab	What it does
1 Generate plot grid	Draws the plot shapefile on an RGB orthomosaic from the 4 trial corners and the layout (no Metashape needed).
2 Edit plot grid	Moves, reshapes, grows/shrinks, widens/lengthens or shifts plots on the RGB orthomosaic; corrects the grid to the drilled rows.
3 IIQ conversion	Converts Phase One IIQ raw files to TIFF, GeoTIFF (georeferenced) or JPEG.
4 Extract plots	Our method: one image per plot cut from the Phase One JPEGs (pixels identical to the photos), or from the orthomosaic.
5 IIQ plot extraction	The same extraction, with each plot developed straight from its IIQ raw file (approach of APPN, The University of Queensland, with our changes).
6 Training tiles	6a: 1 m × 1 m tiles (any size, the same on every plot) from the middle crop rows or the whole plot. 6b: whole rows (all, middle N or chosen rows) and same-size tiles along each row.
7 QA	Checks the plot images against their source, the tiles against the plot images, and the rows; pictures per plot.
8 Acknowledgements	Credits APPN (The University of Queensland) and Phase One for the IIQ plot extraction, and what we changed.

The current grid. The grid saved in tab 1 or tab 2 becomes the current grid: tab 2 opens it for editing, and tabs 4 and 5 use it by default. Each of tabs 4 and 5 can still use any other plots file (Browse).

Coordinate system. All outputs are written in **EPSG:7850 (GDA2020 / MGA zone 50)** by default: the grids of tabs 1 and 2 (reprojected from the orthomosaic's or input file's CRS) and the georeferenced images of tabs 3-5. Each of these tabs has an Output CRS / Save in CRS field for another EPSG code (e.g. 7851 for zone 51). Inputs may be in any CRS. Loading an orthomosaic in tab 1 or 2 sets all these fields to the orthomosaic's CRS (a lon/lat orthomosaic gives its local MGA2020 zone, 7850 at York); you can still change them.

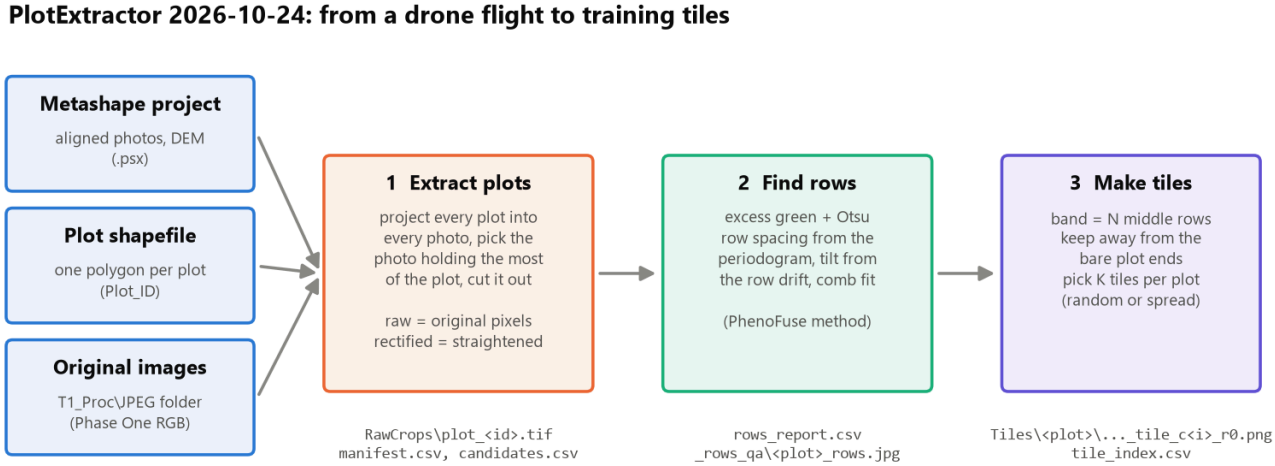

Figure 1. *The core steps (grid, plot images, tiles) and the files each step writes.*

It was built for Phase One RGB cameras (the T1_Proc\JPEG images) but works with any camera that Metashape has aligned. With the recommended settings the plot images are **pixel-for-pixel identical to the original photos** (chapter 10 and the technical report).

2. Quick start: the recommended way

For Phase One RGB images and original-quality plot images plus middle-row training tiles:

1. Start PlotExtractor\PlotExtractor.exe.
2. **Tab 1 or 2:** make the plot grid on the RGB orthomosaic (tab 1), or load your grid in tab 2 and fine-tune it; **Save & use** makes it the current grid.
3. **Tab 4 (Extract plots):** choose the Metashape project (.psx) and, under Raw images folder, the flight's original T1_Proc\JPEG folder; the plot grid is already filled in. Choose an output folder. (For the highest image quality use tab 5 instead, chapter 8.)
4. Set **Margin 0.10 m**, **Format GeoTIFF**, tick **Raw image crop per plot** with style **Masked to plot**, and leave **Convert raw crops to sRGB** off.
5. Set **Limit 3** and press **Run extraction** as a test. Open a few crops in RawCrops\. If they look right, set Limit back to 0 and run all plots.
6. **Tab 6 (Training tiles):** Plot crops folder = <output>\RawCrops; choose a tiles output folder.
7. **Tab 6a:** Tile size **1.0 m**, tick **Tile only the middle 3 rows**, set **sown rows per plot** (5 at York), **Tiles per plot** (e.g. 4), pick **random** or **spread**, keep tiles **0.5 m** from the plot ends.
8. Format **tif**, Bits **same as plot image**, TIFF compression **LZW**, tick **Georeference tiles**, **Link tiles to the source photo / IIQ**, **One folder per plot** and **QA pictures**, press **Make tiles**. Every tile is 1 m × 1 m and the same pixel size on every plot.
9. **Tab 6b (rows):** Rows all, tick whole rows and row tiles (1 m), press **Extract rows**: <plot>\rows\ and <plot>\row_tiles\row<k>\.
10. Check _rows_qa\ (one picture per plot) and rows_report.csv, and run **tab 7 (QA)** on the output.

Recommended settings for Phase One RGB images

Extract plots	
Raw images folder	**the original T1_Proc\JPEG folder of the flight**
Format	**GeoTIFF (geotagged, lossless)**
Raw crop style	**Masked to plot (100% original pixels)**
Convert to sRGB	**off (keep the captured colours)**
Margin around plot	**0.10 m (outer rows whole when the grid is a little off)**
Limit	**3-5 for a test run, then 0 = all plots**
Training tiles	
Plot crops folder	**<output>\RawCrops**
Tile only the middle N rows	**on, N = 3**
Sown rows per plot	**the trial's sown rows (5 at York); 0 = detect**
Square tiles spanning the band	**on**
Tiles per plot	**e.g. 4 (0 = all that fit) · pick: random or spread**
Keep tiles ... m from plot ends	**0.5 m**
Format / folders	**PNG · one folder per plot · QA pictures on**

Figure 2. *Recommended settings at a glance.*

Video: docs\PlotExtractor_Demo.mp4 walks through these steps in the real application (about 7 minutes, narrated; recorded with the earlier tab layout, before the tabs were numbered).

3. Installing and starting

- **Application:** download the release zip from github.com/appndpird/plot-extractor (Releases), unzip it and run PlotExtractor\PlotExtractor.exe. Keep the whole folder together. A console window opens next to the main window and shows progress and errors.
- **From source:** PlotExtractor.bat or python plot_extractor.py with the Python 3.11 that has the Metashape 2.3.1 module, rasterio, Pillow and pyshp.
- **Licence:** tabs 3 (GeoTIFF only), 4 and 5 need a Metashape licence (the floating licence server; VPN if off-site). Tabs 1, 2, 6 and 7 do not.
- **Phase One SDK:** tabs 3 and 5 develop IIQ files with the Phase One ImageSDK. The DPIRD build carries it; otherwise point **Phase One SDK folder** at an SDK obtained from Phase One. The status line under it must say OK.
- **Settings** are remembered in config.json next to the application.

4. Tab 1 · Generate plot grid

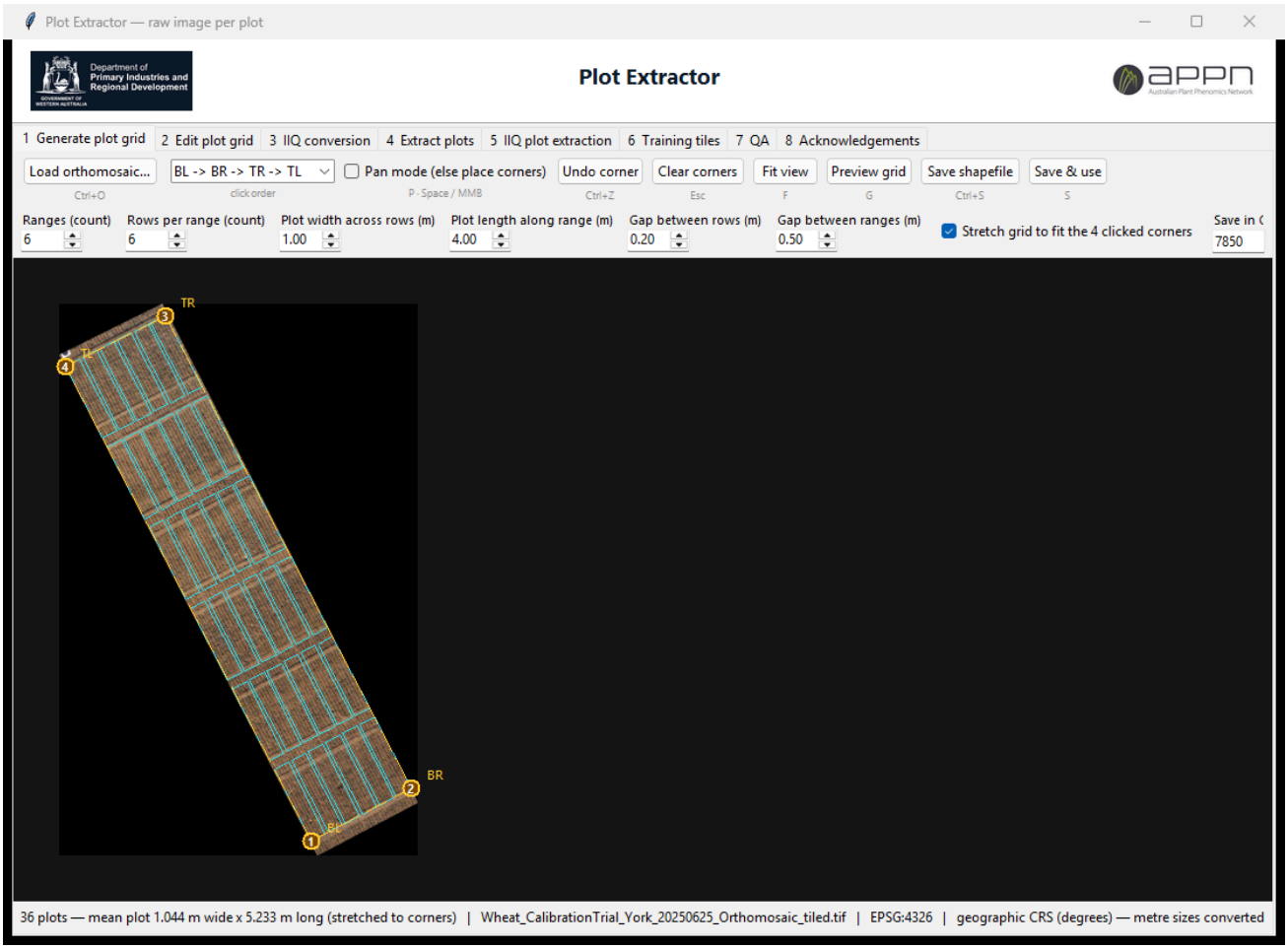

Figure 3. *Tab 1: the 4 trial corners clicked on the RGB orthomosaic and a 6 x 6 grid previewed (York 2025).*

Use it when you do not have a plot shapefile. It needs only an RGB orthomosaic GeoTIFF, no Metashape licence.

1. **Load orthomosaic…** (Ctrl+O). Mouse wheel zooms, Space or middle-drag pans, F fits the view.
2. Click the **4 corners of the trial** in the order shown in the drop-down. Right-click undoes the last corner.
3. Enter **Ranges**, **Rows per range**, plot width and length and the gaps. Keep **Stretch grid to fit the 4 clicked corners** on for rotated trials.
4. **Preview grid** (G) and check the reported plot size, then **Save & use** (S): the shapefile is written in the **Save in CRS** (default EPSG:7850), becomes the current grid and opens in tab 2.

Fine-tune the grid in **tab 2** (chapter 5).

5. Tab 2 · Edit plot grid

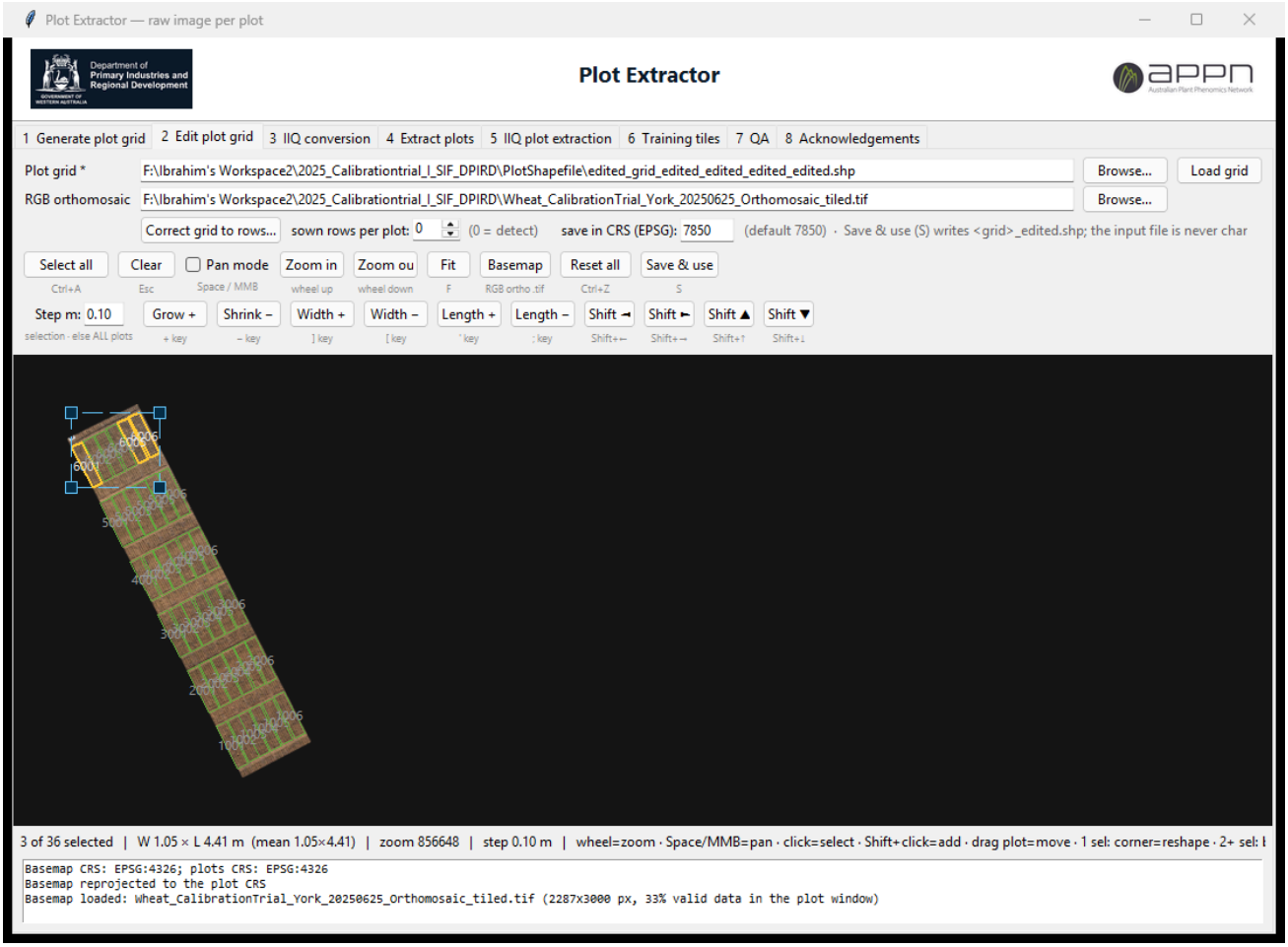

Figure 4. *Tab 2: the York grid on the RGB orthomosaic, three plots selected (blue handles resize them together).*

Edits a plot grid on the RGB orthomosaic. The file you load is never changed: Save & use writes <grid>_edited.shp next to it and makes that the current grid for tabs 4 and 5.

1. **Plot grid:** filled with the current grid (from tab 1, or the last grid you saved); Browse for another and press **Load grid**.
2. **RGB orthomosaic:** the trial's orthomosaic GeoTIFF; it is drawn under the plots (any CRS, reprojected to the plots). **Basemap** on the toolbar loads another one.
3. **Select:** click a plot, Shift/Ctrl-click to add, drag a box around plots, Ctrl+A for all. With nothing selected the tools act on **all** plots.
4. **Edit:** drag plots to move them; with one plot selected drag its corners; with 2+ selected drag a blue handle to resize them together. The buttons change the selection by the **Step** (m): Grow/Shrink, Width ±, Length ±, Shift ◄ ► ▲ ▼ (keys + –] [' ; and Shift+arrows).
5. **Save & use** (S). The edited grid is written in the **save in CRS** (default EPSG:7850). Reset all (Ctrl+Z) goes back to the loaded grid.

Correct grid to rows… measures every plot against the orthomosaic and centres it on its drilled rows (set the sown rows per plot, or 0 to detect them). It writes <grid>_refit.shp with an audit CSV and asks whether to use it. It refuses when the rows run continuously across plots, because then the imagery has no plot edge to fit.

The editor is for RGB orthomosaics; the bottom box shows its messages.

6. Tab 3 · IIQ conversion

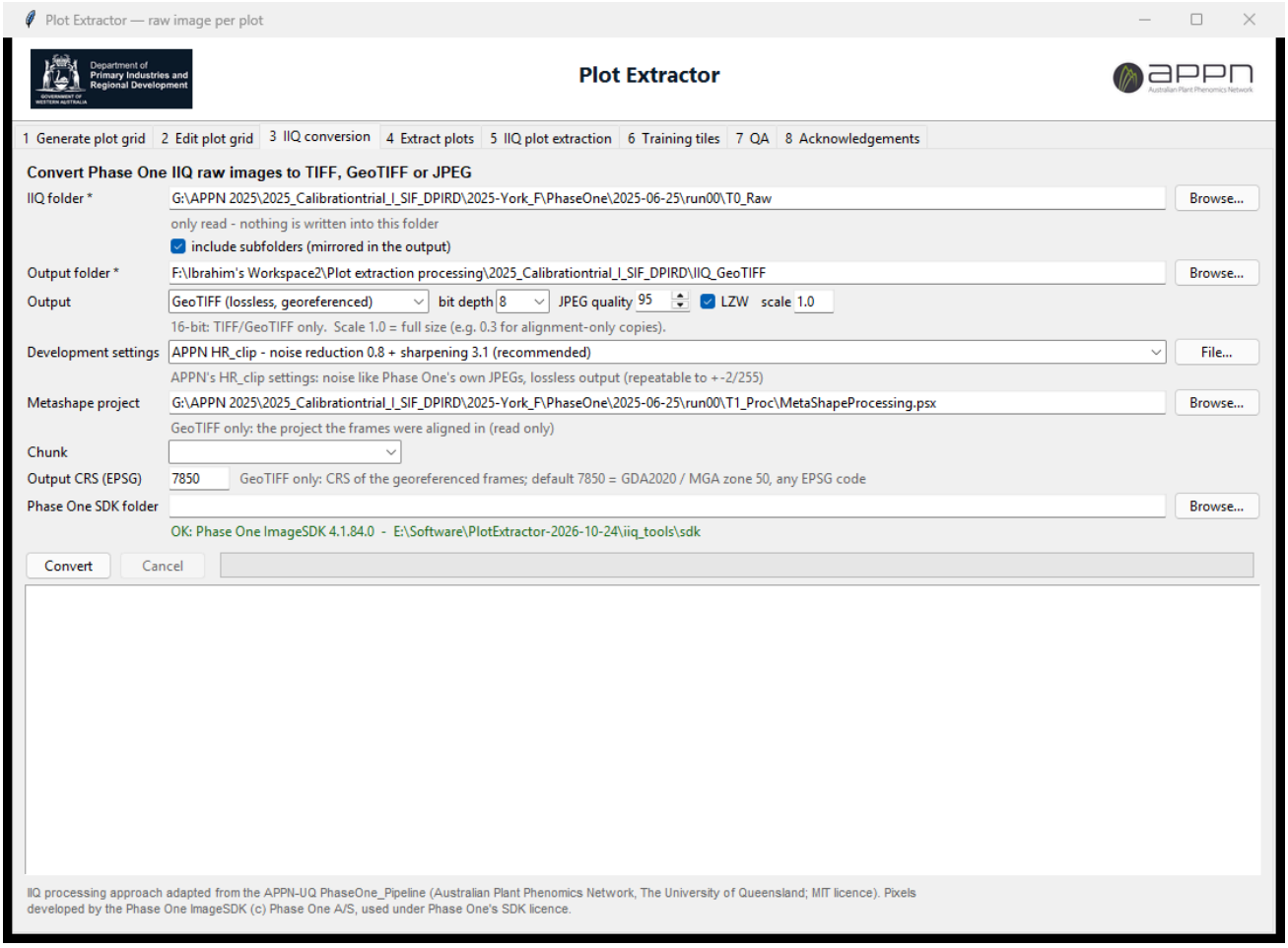

Figure 5. *Tab 3: all IIQ files of a flight to georeferenced GeoTIFFs with the APPN HR settings.*

Converts Phase One IIQ raw files to images with the Phase One ImageSDK. The IIQ folder is only read; an output folder inside it is refused.

Field	What it is	Recommended
IIQ folder	The flight's T0_Raw folder (subfolders optional, mirrored in the output)	T0_Raw
Output	TIFF (8 or 16 bit, LZW), GeoTIFF (georeferenced) or JPEG; scale 1.0 = full size	GeoTIFF, 8 bit, 1.0
Development settings	APPN HR_clip (noise reduction 0.8, sharpening 3.1), Neutral (no processing), APPN template, or a Phase One settings file	APPN HR_clip
Metashape project / Chunk	GeoTIFF only: the project the photos were aligned in. Each frame gets ground control points from its aligned camera and the DEM and an affine georeference (a lon/lat project is written in the local UTM/MGA zone)	The flight's .psx
Output CRS (EPSG)	GeoTIFF only: CRS of the georeferenced frames	7850 (default)
Phase One SDK folder	Blank = the SDK shipped with the app	Blank

Each output keeps the camera EXIF/GPS. conversion_log.csv lists every file, and IIQ_settings_used.json the settings. Checked on York 2025: a converted frame lies within 1 cm of the orthomosaic.

7. Tab 4 · Extract plots

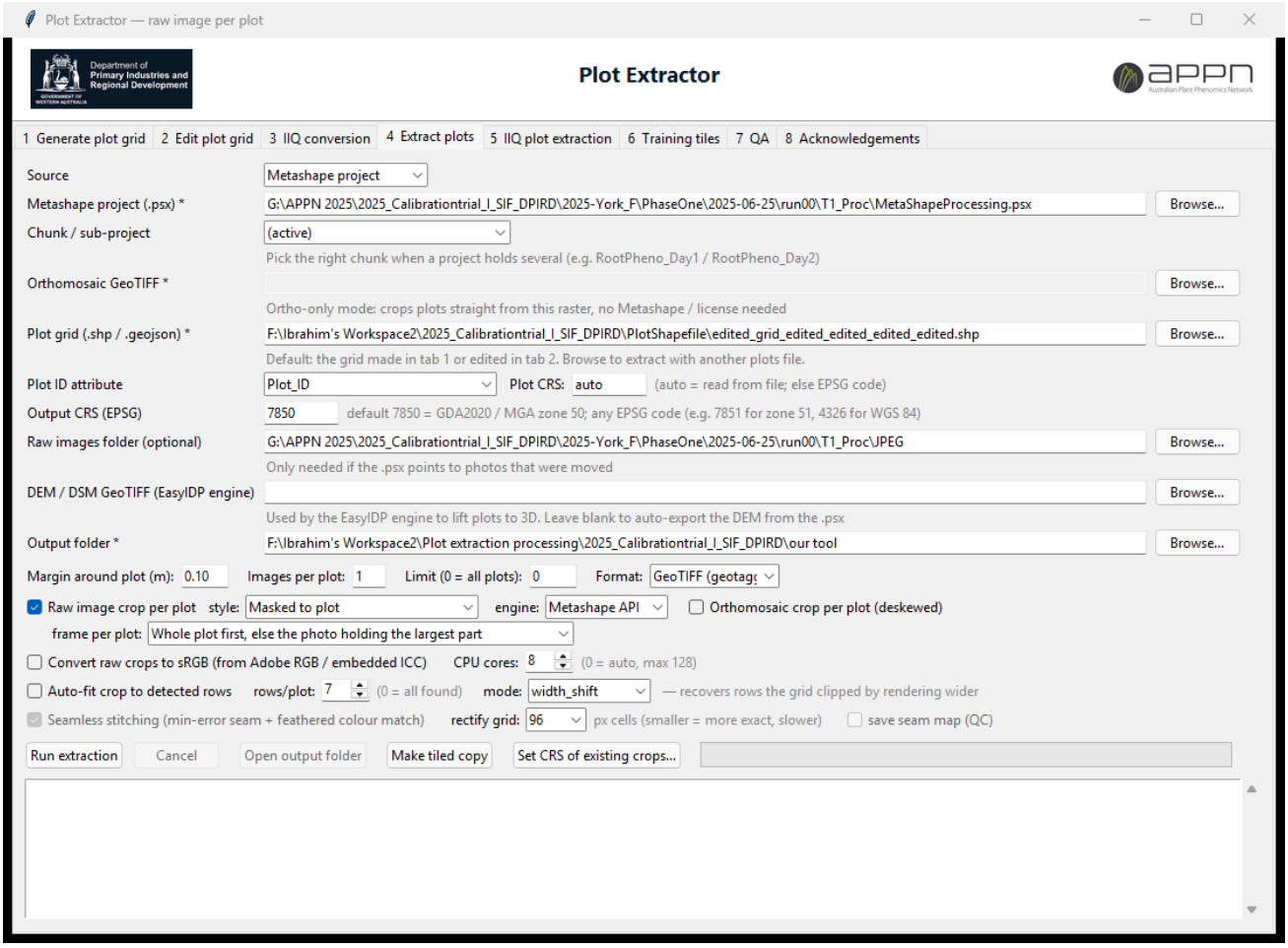

Figure 6. *Tab 4, Extract plots, with the recommended settings for the York calibration trial.*

7.1 The fields

Field	What it is	Recommended
Source	Metashape project (raw + ortho crops) or Orthomosaic GeoTIFF (ortho crops only, no licence)	Metashape project
Metashape project (.psx)	The flight, aligned and georeferenced, with a DEM	The run's .psx in T1_Proc
Chunk / sub-project	When a project holds several chunks	The chunk of this flight
Plot grid	Shapefile or GeoJSON, one polygon per plot, any CRS. Filled with the current grid from tab 1/2; Browse for another file	The current grid
Plot ID attribute	Field used to name the files (plot_<id>.tif)	Plot_ID (picked automatically)
Output CRS (EPSG)	CRS of the GeoTIFF georeference of every crop	7850 (default)

Field	What it is	Recommended
Raw images folder	Where the original photos are. Preferred over the paths stored in the project	T1_Proc\JPEG of the flight
Output folder	Gets RawCrops\, manifest.csv, candidates.csv	A new folder per flight
Margin around plot (m)	Extra ground on every side of the plot	0.10 (0.2 if the grid is far off)
Images per plot	Extra views per plot (plot_<id>_2 ...)	1
Limit	Only the first N plots, for a test run	3-5 to test, then 0 = all
Format	GeoTIFF (lossless, geotagged), PNG, TIFF, JPEG (lossy)	GeoTIFF
Raw image crop per plot · style	How each plot is cut (section 7.2)	On · Masked to plot
Orthomosaic crop per plot	A straightened orthomosaic crop as well	Optional
Convert raw crops to sRGB	Converts the Adobe RGB colours of Phase One JPEGs to sRGB	Off for original pixels
CPU cores	Parallel image work; each core holds a full frame (~370 MB)	0 (auto) or 8
Frame per plot	Whole plot first, else the photo holding the largest part (or the older nadir-tier order)	Whole plot first
Auto-fit crop to detected rows	Snaps the crop to the detected rows. Needs an upright crop: with Native, Masked or Bounding box it switches the style to Rectified (resampled, may stitch photos). With rows/plot 0 it keeps every row it finds, including the outer rows of the neighbouring plots	Off. If used: rows/plot = the sown rows (e.g. 5)
Seamless stitching / rectify grid	Only for rectified crops	Defaults

Plot only, as drawn in the shapefile: Masked to plot (or Native), margin 0, Auto-fit off, sRGB off. The crop then follows the plot grid exactly, so the grid decides what is in the image: two grids of the same trial (e.g. a refit grid) give crops shifted by the same distance.

7.2 Raw crop styles: which one to use

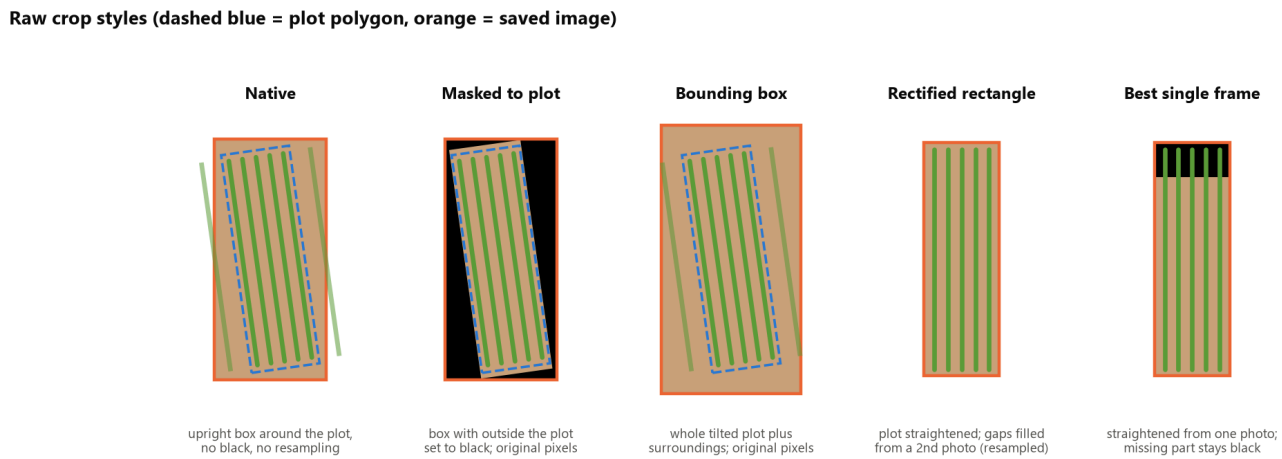

Figure 7. *The raw crop styles on a plot that is slightly tilted in the photo.*

Style	Pixels	Use it when
Masked to plot	Original, unchanged. Outside plot + margin set to black	**Recommended.** You need 100% original quality (training data, counting).
Native (no resample, no black)	Original, unchanged. Upright box, no black	You prefer no black corners and accept some neighbouring ground.
Bounding box	Original, unchanged. Whole tilted plot + surroundings	You want extra context around the plot.
Auto-straighten tilted	Original if the plot is upright, resampled if tilted > 3°	A mixed set where most plots are upright.
Rectified rectangle	Resampled (~5% lower resolution). Gaps filled from a 2nd photo	Plots must be upright and whole, e.g. plot-level measurements.
Best single frame	Resampled; what one photo misses stays black	No stitching allowed at all.

Why not always rectified? Straightening a plot means redrawing every pixel. The result looks the same but is not the original image (about 1.05 original pixels per output pixel). Masked to plot copies the pixels from the photo unchanged.

7.3 What you get

- RawCrops\plot_<id>.tif: one image per plot (GeoTIFF with ground control points, the photo's EXIF and colour profile).
- manifest.csv: one row per plot: source photo, crop size, raw_covered_pct (how much of the plot the image holds), frames_used, raw_style and the pixel box in the photo (src_x0 … src_y1).
- candidates.csv: every photo that sees each plot, with its coverage, so you can see the alternatives.

Figure 8. *A raw (masked) and a rectified crop of the same plot, and the raw crop against the original photo: identical pixels.*

7.4 Check the coverage

Sort manifest.csv by raw_covered_pct. Below 100 means no single photo holds the whole plot, so the raw image is cut at the photo edge. At York this happened to 4 of 36 plots (0.4-3.9% missing). For those plots use the rectified image if you need the whole plot, or see section 12.4 for the flight-planning fix.

8. Tab 5 · IIQ plot extraction

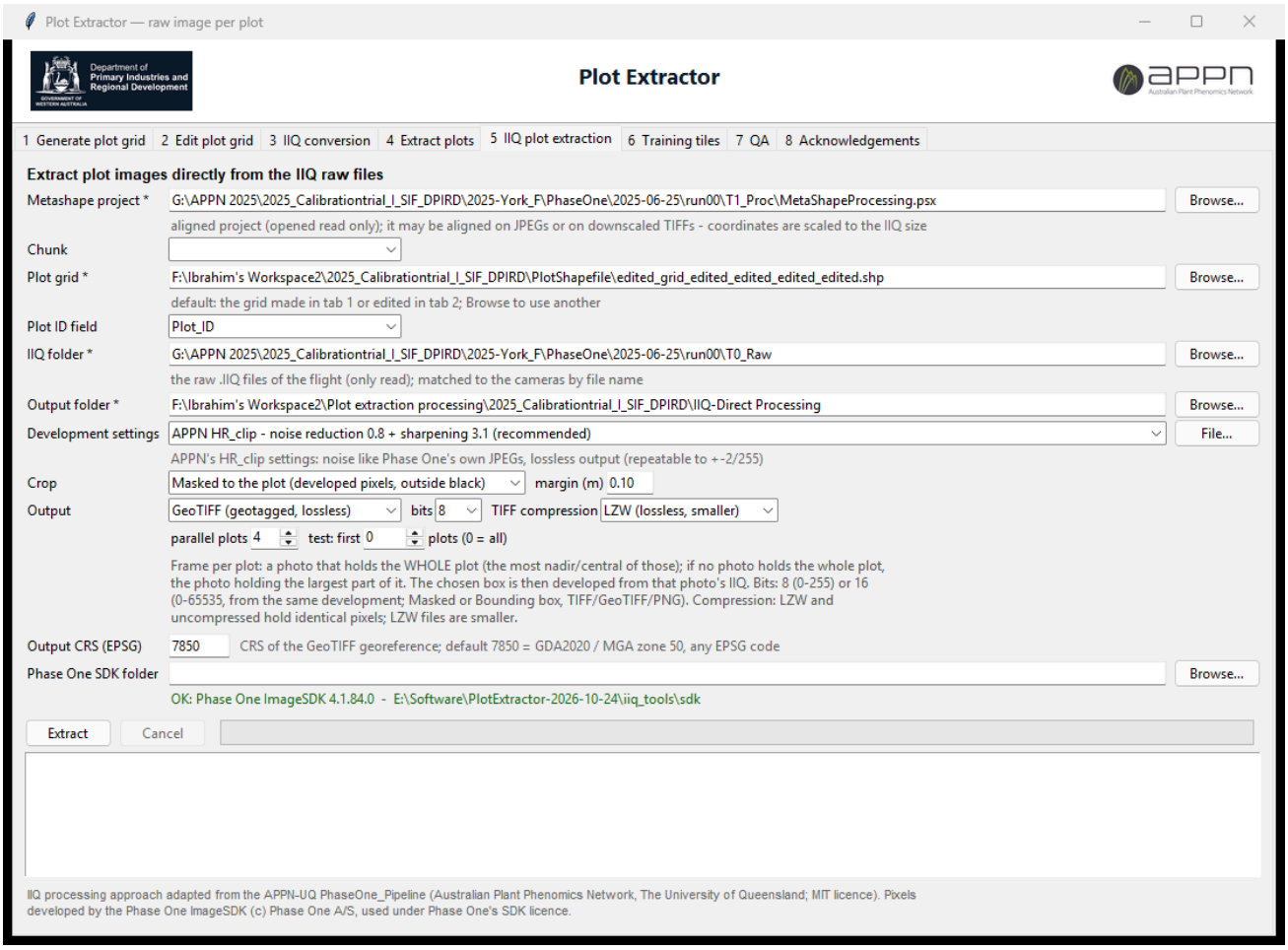

Figure 9. *Tab 5: the York plots developed from the T0_Raw IIQ files with the APPN HR settings.*

The same extraction as tab 4, but each plot's box is developed straight from its IIQ raw file instead of cut from the JPEG. It gives the highest image quality (less noise, more fine detail, no JPEG compression).

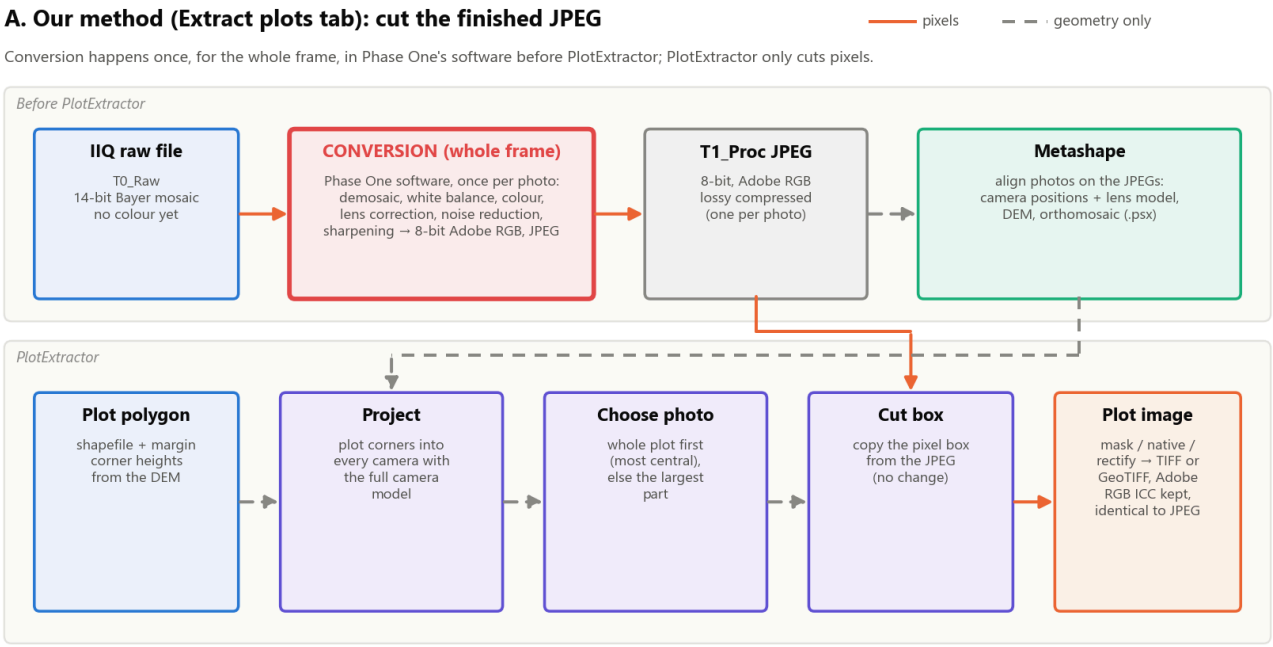

Figure 10. *Tab 4: the conversion is done once, before PlotExtractor, by Phase One's software; the plot is cut from the JPEG unchanged.*

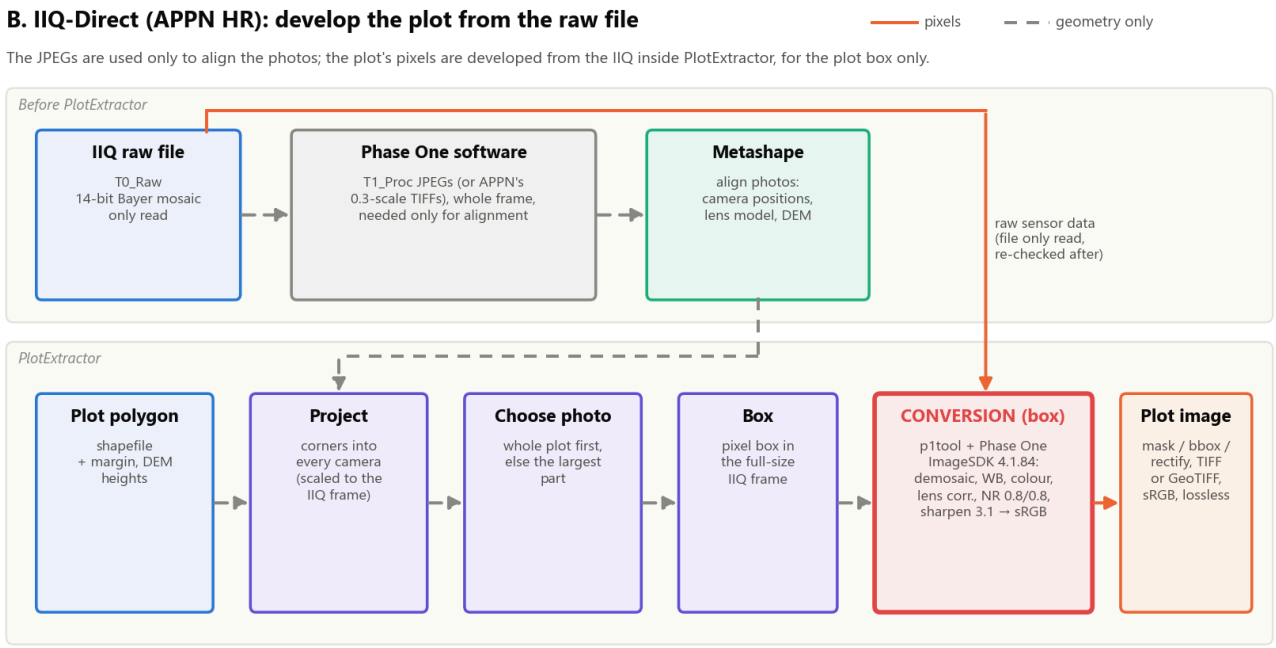

Figure 11. *Tab 5: the JPEGs only give the geometry; the plot box is developed from the IIQ by the Phase One ImageSDK.*

1. **Metashape project:** the aligned flight (aligned on the JPEGs or on downscaled TIFFs; coordinates are scaled to the IIQ size).
2. **Plot grid:** filled with the current grid; Browse for another file. Plot ID field: Plot_ID.
3. **IIQ folder:** the flight's T0_Raw folder (only read; matched to the cameras by file name). Output folder: a new folder.
4. **Development settings:** APPN HR_clip for the best-looking images; Neutral when the least-processed pixels matter (e.g. spectral measurements; noisier). Crop: Masked to the plot, margin 0.10 m.
5. **Output:** GeoTIFF, **bits 8**, **TIFF compression LZW** (section 8.1).
6. Test with **first 3 plots**, check the crops, then 0 = all and **Extract**.

	Tab 4 (JPEG)	Tab 5 (IIQ, APPN HR)
Pixels	Identical to the T1_Proc JPEG	Developed from the raw sensor data
Quality (York 2025, 29 plots)	noise 3.55, detail/noise 5.4	noise 3.13, detail/noise 7.4
Colour	Phase One's JPEG colours (Adobe RGB)	SDK colours (sRGB), a little less green
Speed / needs	Fast; Metashape licence	Slower; also the IIQ files and the Phase One SDK
Repeatable	Bit for bit	Within 1-2/255 (noise reduction); Neutral bit for bit

Use one source per trial and season. Vegetation indices from tab 4 and tab 5 images are not interchangeable (York: green cover 3.0% vs 2.3%, same plot ranking).

8.1 Bits and compression

Option	What it does	Recommended
bits 8	0-255 per channel, as the JPEGs and most training tools expect	Yes
bits 16	0-65535 per channel from the same development (divided by 257 it matches the 8-bit image to 1/255). Masked or Bounding box style; GeoTIFF, TIFF or PNG (not JPEG, not Rectified). Twice the file size; PIL reads 16-bit RGB TIFFs wrongly - use GDAL/rasterio or tifffile	Only for radiometric or exposure work
TIFF compression LZW	Lossless; smaller files (about 10% for camera images)	Yes
TIFF compression None	Uncompressed; the same pixel values as LZW	Only if a tool cannot read LZW

All TIFF and GeoTIFF files are written with GDAL; the developed values are never changed after the SDK (masking only sets the pixels outside the plot to 0). QA (tab 7) re-develops every crop at its own bit depth.

The frame for each plot is a photo holding the whole plot (most central of those) or, if none does, the one holding the largest part. The manifest records pixel_source = iiq; IIQ_settings_used.json and IIQ_run.json record the settings and the SDK; every IIQ used is re-checked unchanged at the end.

8.2 How a plot is made from the IIQ, step by step

A JPEG in T1_Proc\JPEG has already been developed once by Phase One's software, for the whole photo: 8 bit, Adobe RGB, lossy compression. The IIQ in T0_Raw holds the raw 14-bit sensor data from before that. Tab 5 develops each plot from the IIQ itself; the JPEGs are only used by Metashape to align the photos (geometry, no pixels). The example is York calibration trial plot 6006 (25 June 2025).

Step 1 · Find the plot in every photo

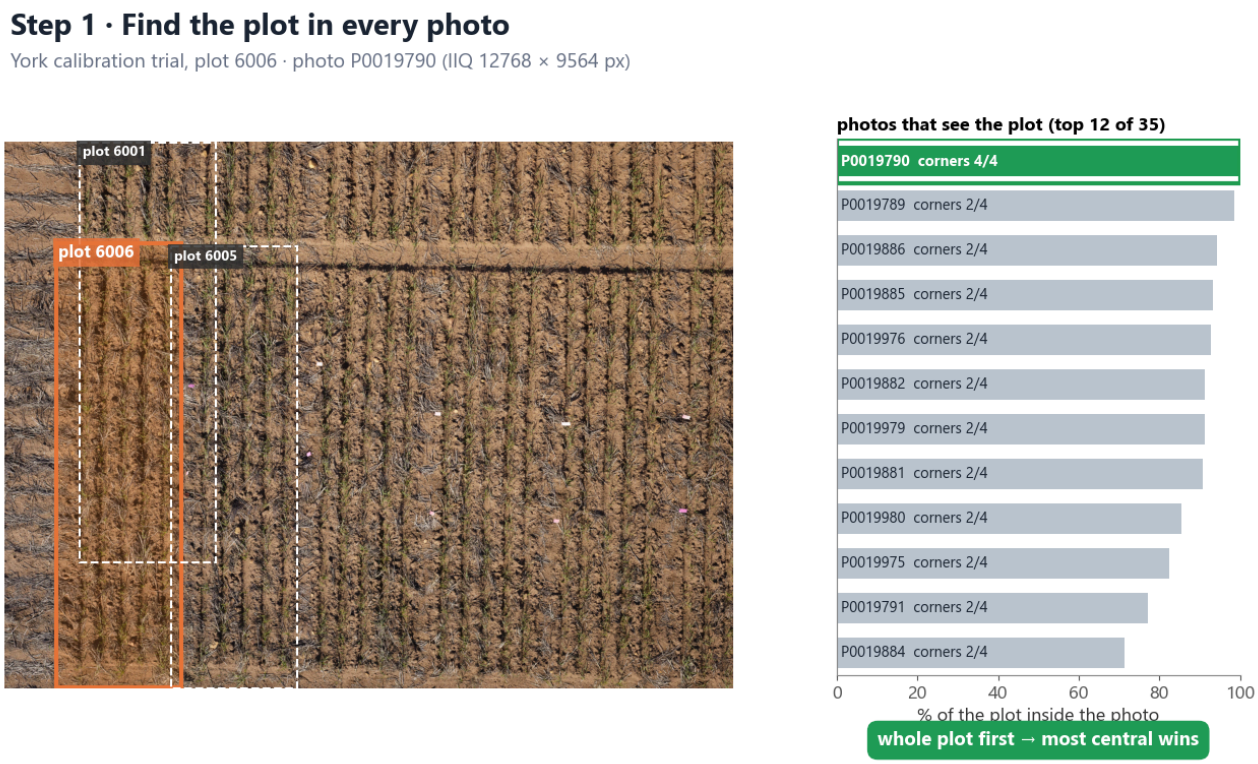

Figure 12. *Plot 6006 in photo P0019790 (left) and the photos that see it (right). Only P0019790 holds all four corners, so it is used.*

The plot polygon plus the margin is placed on the DEM and projected into every photo with Metashape's camera model, lens distortion included (coordinates are scaled to the IIQ size when the project was aligned on smaller images). For plot 6006, 35 photos see the plot. **Photos holding the whole plot come first, and among those the most central one wins** (it looks most straight down). Only when no photo holds the whole plot is the photo holding the largest part used. The candidates are listed in candidates.csv.

Step 2 · Develop only the plot's box

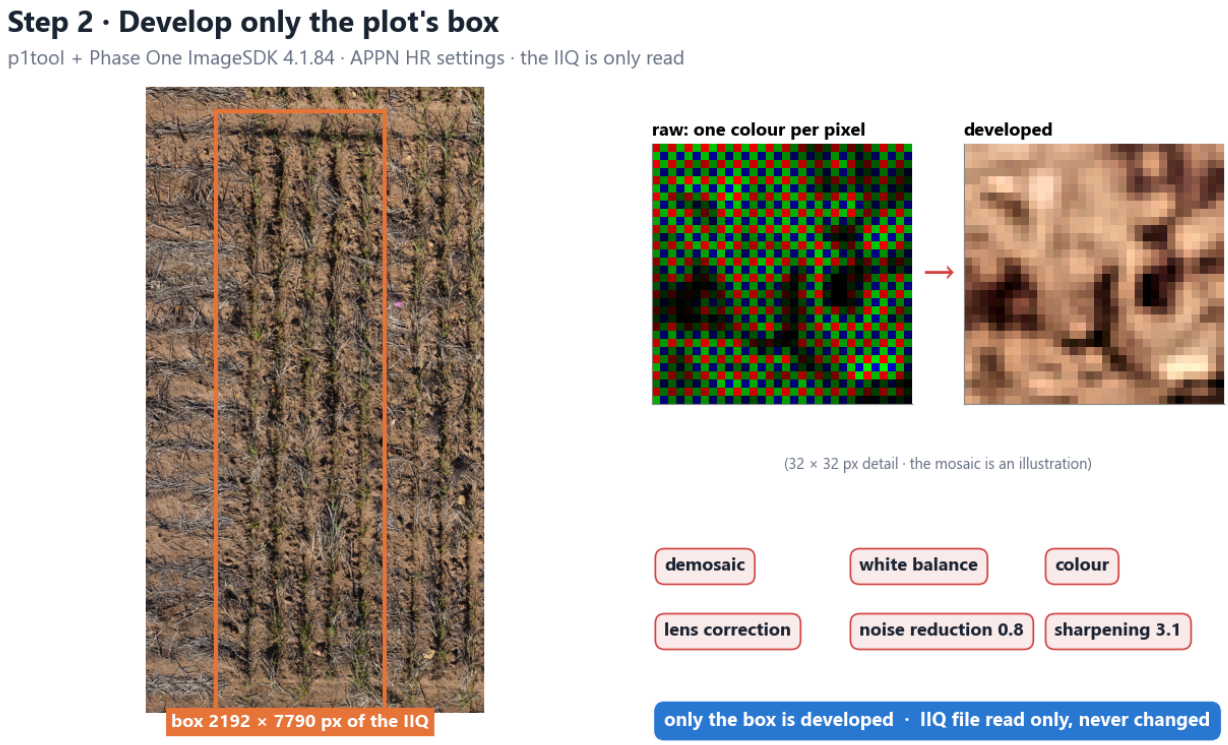

Figure 13. *The plot's box in the chosen IIQ (left) is developed by the Phase One ImageSDK. Right: a 32 x 32 px detail as the sensor saw it (one colour per pixel, an illustration) and developed.*

The plot's pixel box in that photo is passed to p1tool.exe, which uses the Phase One ImageSDK to develop only that box: rebuild full colour from the sensor's colour mosaic (demosaic: each sensor pixel saw only red, green or blue), then white balance, colour, lens correction, noise reduction (0.8) and sharpening (3.1) with the APPN HR settings, at 8 or 16 bit. Only the box is developed, so it is quick; the IIQ file is only read, never changed.

Step 3 · Mask, georeference, save

Figure 14. *The masked plot (left) and the same GeoTIFF in map coordinates, EPSG:7850 (middle).*

Pixels outside the plot polygon are set to black; no other pixel is changed. The image is written with GDAL as a GeoTIFF in GDA2020 / MGA zone 50 (EPSG:7850), with an affine georeference fitted through the plot's ground control points, so it opens in the right place in QGIS. Its tags name the IIQ (SOURCE_IMAGE) and manifest.csv records the photo and the pixel box. 8 or 16 bit and LZW or no compression: section 8.1.

9. Tab 6 · Training tiles

Tab 6 has two sub-tabs. **6a Plot tiles** cuts square tiles of a fixed ground size (1 m × 1 m by default) from every plot image. **6b Rows and row tiles** saves whole crop rows (all of them, the middle N, or the rows you list) and cuts tiles of a fixed ground size along each row. In both, every tile has the same ground size **and** the same pixel size on every plot and every row, and each run records its settings in PlotExtractor_recipe.json (section 9.4).

9.1 Tab 6a · Plot tiles of a fixed ground size

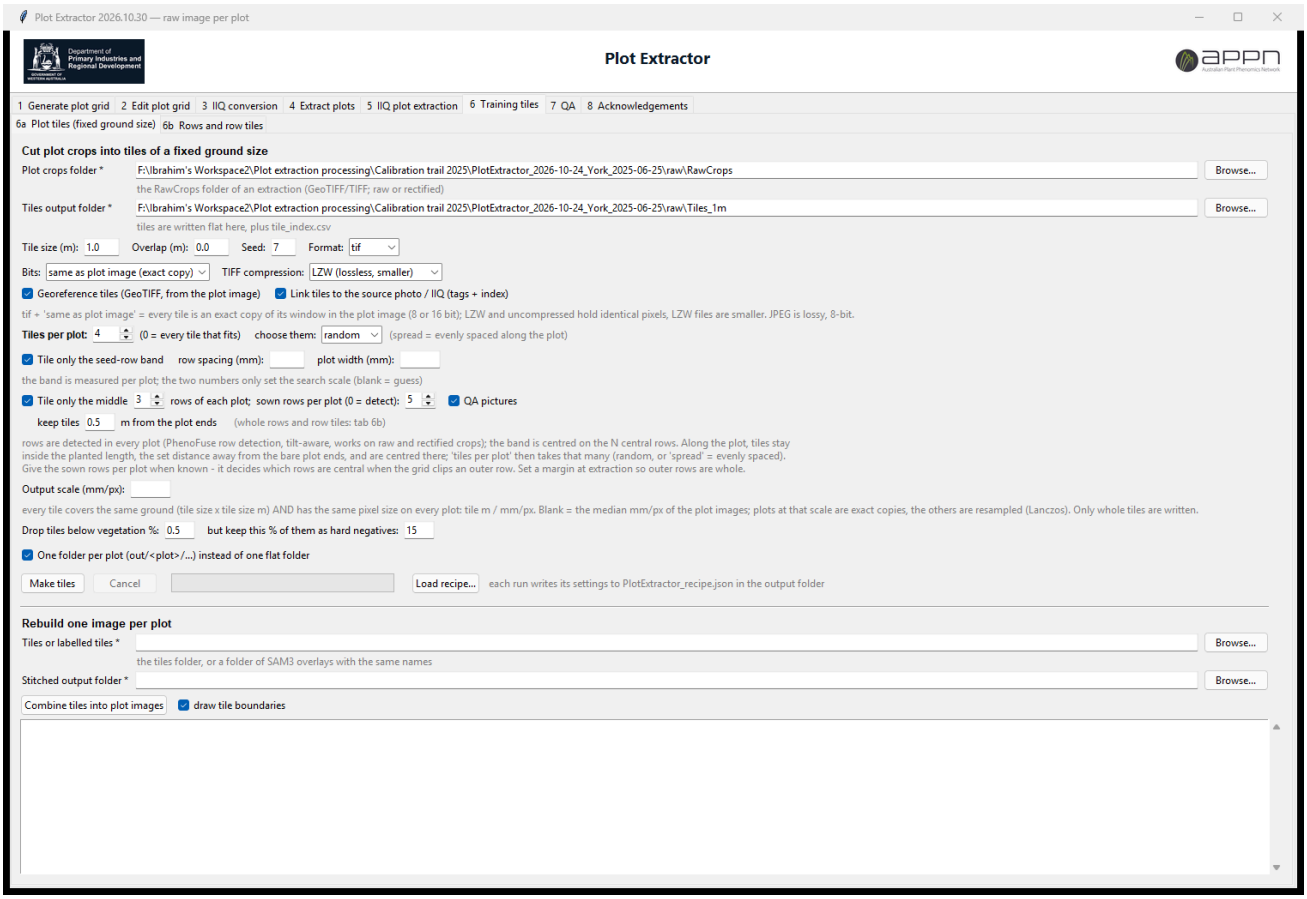

Figure 15. *Tab 6a set up for 1 m × 1 m tiles on the middle 3 rows, 4 per plot, 0.5 m away from the plot ends.*

1. **Plot crops folder:** the RawCrops folder of an extraction, raw or rectified (GeoTIFF: the ground scale comes from its georeference).
2. **Tiles output folder:** a new folder; tiles plus tile_index.csv and the recipe are written there.
3. **Tile size (m):** 1.0 for 1 m × 1 m tiles; **Overlap (m)** 0 for tiles laid end to end.
4. Tick **Tile only the middle N rows**, set N (3) and **sown rows per plot** (5 at York), or untick it for a grid over the seed-row band or the whole crop.
5. Set **Tiles per plot** (0 = every tile that fits), **choose them** random or spread, and **keep tiles … m from the plot ends** (0.5).
6. Leave **Output scale (mm/px)** blank (the plots' median) or set it to make tiles of several runs the same pixel size.
7. Format **tif**, Bits **same as plot image**, **LZW**, tick **Georeference tiles**, **Link tiles to the source photo / IIQ**, **One folder per plot**, **QA pictures**, then **Make tiles**.

How the size is kept the same. A plot image's ground scale (mm per pixel) comes from its georeference and differs slightly from plot to plot (each raw crop is from a different photo). The tool reads every plot's scale first

and takes one **output scale** for the run: the scale of the plot nearest the median, or the value you set. A tile is then tile size / output scale pixels square on every plot (York raw: 1 m / 0.568 mm = 1761 px). On each plot the window is cut at that plot's own scale and, when it differs, resampled (Lanczos) to the tile size; plots at the output scale are exact copies. Only whole tiles are written: a plot shorter or narrower than one tile gets none (the log says so). The log ends with a size check, e.g. pixel sizes [1761] (all the same).

Option	What it does	Recommended
Tile size (m)	Side of every tile on the ground	1.0
Overlap (m)	Overlap between neighbouring tiles	0
Output scale (mm/px)	Pixel size of the tiles; blank = the scale of the plot nearest the median. Set it to match tiles from different runs or trials	Blank, or a fixed value across trials
Tile only the middle N rows	Tiles centred on the band of the N central crop rows of each plot	On, N = 3
sown rows per plot	How many rows were sown. Decides which rows are central when the grid clips an outer row	The trial's value (5 at York)
keep tiles … m from the plot ends	Tiles stay this far inside the plot ends (after trimming to the planted length); end to end and centred in what is left	0.5 m
Tiles per plot / choose them	0 = every tile that fits; otherwise random (Seed makes it repeatable) or spread (evenly along the plot)	As needed
Tile only the seed-row band	Grid mode (middle rows off): only the band holding the rows	On
Format, Bits, TIFF compression	tif/png lossless, jpeg lossy 8-bit; same as plot image = exact values; LZW and None hold identical pixels	tif, same, LZW
Georeference tiles	TIFF tiles are GeoTIFFs: the plot image's CRS and transform moved (and scaled) to the tile	On
Link tiles to the source photo / IIQ	Tags and tile_index.csv columns: source photo or IIQ and the tile's pixel box there	On
Drop tiles below vegetation %	Grid mode filter, keeping some empty tiles as hard negatives	Default
One folder per plot	out\<plot>\… instead of one flat folder	On

Exact copies. tile_index.csv has a resampled column: False = the tile holds exactly the values of its window in the plot image (QA, tab 7, checks this pixel by pixel). Tiles of plots at another scale are resampled to the common size, which is what makes all tiles the same size; their colour is unchanged.

9.2 Tab 6b · Whole rows and row tiles

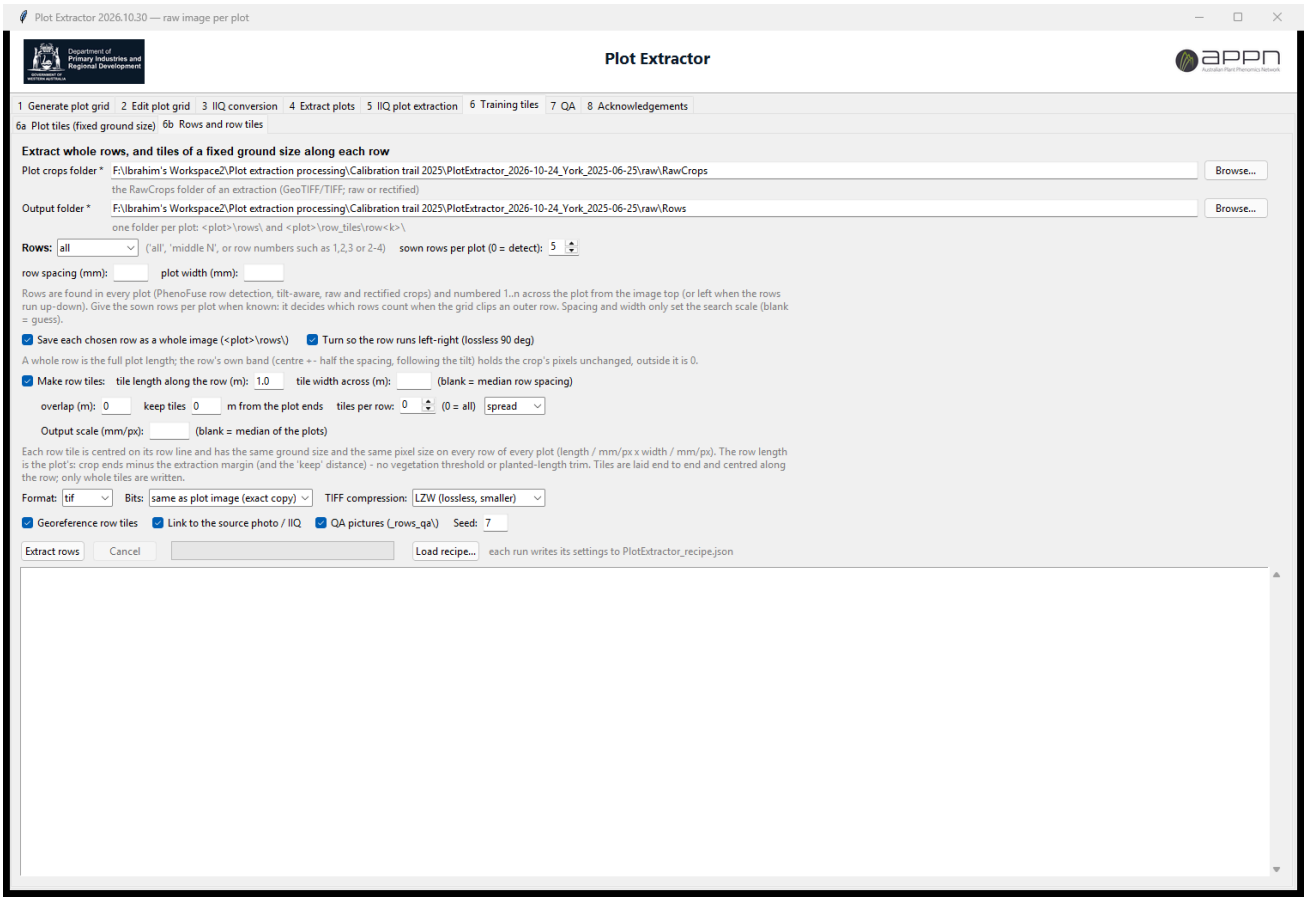

Figure 16. *Tab 6b: all rows of every plot as whole images, plus 1 m row tiles.*

1. **Plot crops folder:** <output>\RawCrops; **Output folder:** a new folder (e.g. <output>\Rows).
2. **Rows:** all, middle 3, or row numbers such as 1,2,3 or 2-4. Rows are numbered 1..n across the plot from the image top (or from the left when the rows run up-down). Set **sown rows per plot** when known.
3. Tick **Save each chosen row as a whole image** and **Turn so the row runs left-right**.
4. Tick **Make row tiles**: **tile length along the row** 1.0 m; **tile width across** blank (= the median row spacing measured over all plots, 0.215 m at York) or a value; overlap 0; keep tiles 0 m from the plot ends; tiles per row 0 (all).
5. Format, bits, compression, georeference and link as in 6a; then **Extract rows**.

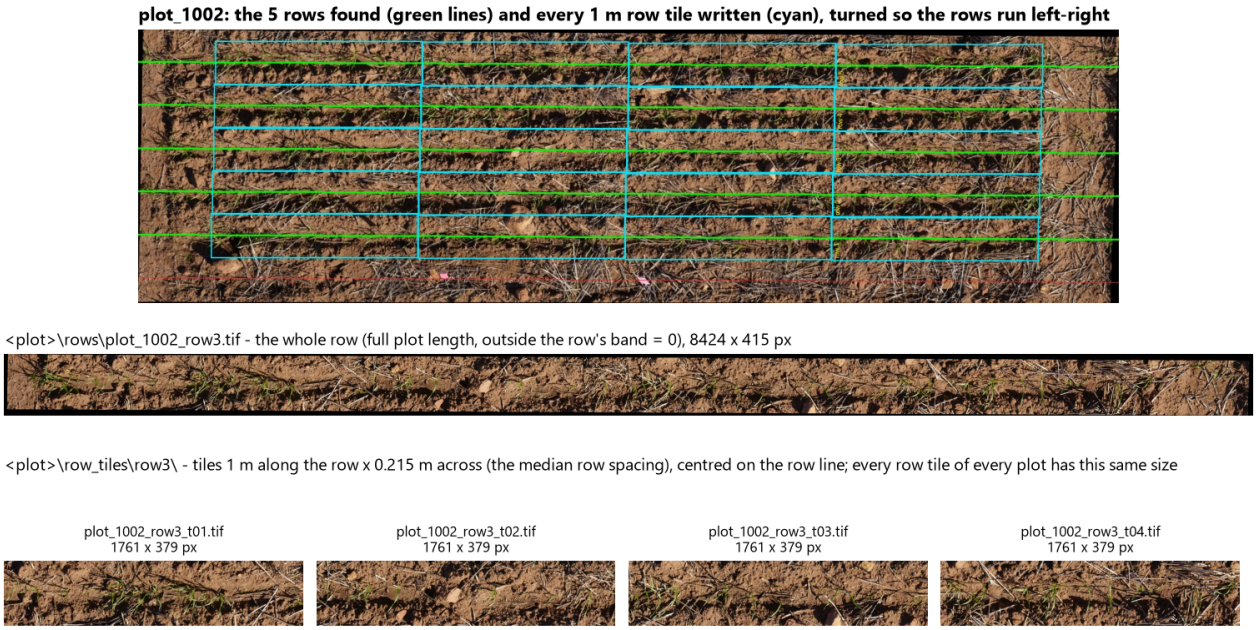

Figure 17. *York plot 1002 (raw crop, rows run up-down in the photo, turned): the 5 rows and every 1 m row tile, one whole row, and the 4 tiles of row 3. All row tiles of all plots are 1761 × 379 px.*

The row's length. A row tile run covers the row inside the plot: the crop ends minus the extraction margin (from manifest.csv) and minus the optional 'keep' distance. There is **no vegetation threshold** and no trimming to the planted length, so gaps and bare stretches of a row are kept. The tiles are laid end to end (with the overlap) and centred in that length; only whole tiles are written (York: 4.2-4.4 m of row gives 4 tiles of 1 m).

Same size everywhere. As in 6a, one output scale is used for the run, so every row tile is length / scale × width / scale pixels on every row of every plot. Each tile is centred on its row's line at the tile's middle (the line follows the tilt). With **Turn** on, plots whose rows run up-down are turned 90° (lossless, georeference turned with them), so all tiles are landscape with the row running left-right.

Folder structure:

Path	Contents
<out>\<plot>\rows\<plot>_row<k>.tif	The whole row k: full plot length, the crop's pixels inside the row's band (centre ± half a spacing, following the tilt), 0 outside
<out>\<plot>\row_tiles\row<k>\ <plot>_row<k>_t<j>.tif	Row tile j of row k (1 = first along the row)
<out>\rows_index.csv	Per whole row: file, plot, row number, of rows, middle, contrast, centre and width (px), tilt, box in the crop, turned
<out>\row_tiles_index.csv	Per row tile: plot, row, tile number, box in the crop (x0, y0, w, h), output size, turned, resampled, plot and output mm/px, tile length and width (m), vegetation %, source photo link
<out>\rows_report.csv	Per plot: rows found, rows used, spacing (mm), tilt, mm/px, row length (m), rows saved, row tiles, or why it was skipped
<out>_rows_qa\<plot>_rows.jpg	The plot with its rows (green) and every row tile (cyan)
<out>\PlotExtractor_recipe.json	The settings of this run (and of the extraction the crops came from)

Option	What it does	Recommended
Rows	all, middle N, or a list (1,2,3 / 2-4). A listed row the plot does not have is skipped and reported	all
sown rows per plot	Rows sown per plot (0 = the run of rows found)	The trial's value
Save each chosen row as a whole image	<plot>\rows\ images	On
Turn so the row runs left-right	90° lossless turn for plots whose rows run up-down	On
tile length along the row (m)	Length of every row tile	1.0
tile width across (m)	Width of every row tile; blank = the median row spacing of the plots	Blank
overlap (m), keep tiles … m from the plot ends	Overlap between row tiles; distance kept from the plot ends	0, 0
tiles per row	0 = all that fit; else that many, spread or random	0
Output scale (mm/px)	As in 6a	Blank

9.3 The rows numbering

The rows are those of the row detection (9.5): with sown rows per plot set, the block of that many neighbouring rows with the highest contrast; otherwise the run of rows found. Row 1 is the one nearest the image top (rows running left-right) or the left edge (rows running up-down) of the plot image, before any turning. 'middle 3' takes the 3 central rows of that block, the same rows tab 6a tiles.

9.4 The recipe file

Every run writes PlotExtractor_recipe.json in its output folder, one section per step:

- plot_extraction: tab 4 or tab 5 - project, plots file, source images, margin, style, format, every option (written when the run starts).
- plot_tiles: tab 6a; rows_and_row_tiles: tab 6b - every field, the values passed to the run and the results (tiles, plots, tile size in pixels, output scale, row width).

A tiles run also copies the plot_extraction section of the run its crops came from (the recipe next to RawCrops), so the recipe in a tiles folder records the whole chain. **Load recipe…** on tab 4, 6a and 6b fills the tab's fields from a recipe file, to repeat a run or apply the same settings to another trial.

9.5 How the rows are found

The rows are found in every plot image by row_detect.py (the PhenoFuse 4.0 row method, adapted to raw photos, where the rows are not always parallel to the image edge). Six steps: a greenness value per pixel, a threshold that separates plants from soil, a profile across the plot, the row spacing, the tilt, and a comb fitted to the rows. The figures show each step on OzBarley 2025 plot 1001 (a young crop, rows tilted 2.8° in the photo, 5 sown rows). The numbers are those the tool computed for this plot.

Step 1 · Greenness of every pixel (excess green)

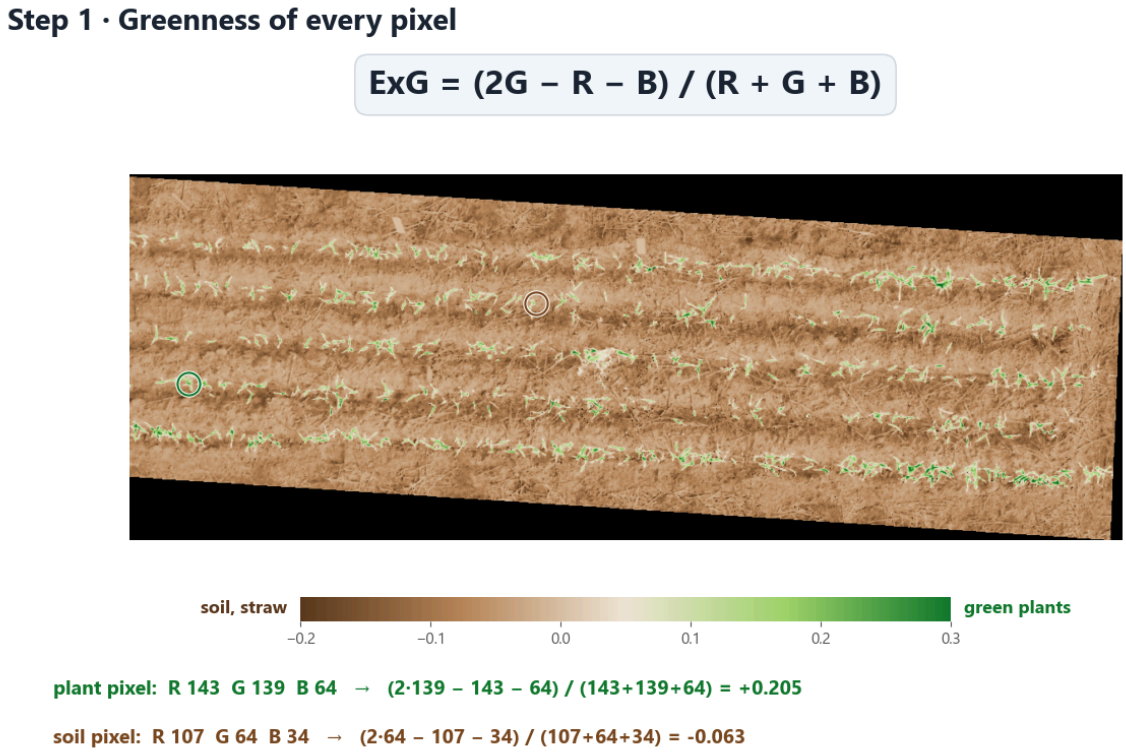

Figure 18. *The plot image turned into excess green: plants high (green), soil and straw low (brown). Two example pixels with their calculation.*

Every pixel gets **ExG = (2G − R − B) / (R + G + B)**. Dividing by R + G + B removes brightness, so a shaded leaf and a sunlit leaf score about the same. Green plants score high (about 0.1 to 0.3); soil and straw about 0 or below. Black pixels (outside a masked plot) are ignored.

Step 2 · Plant or soil: the Otsu threshold

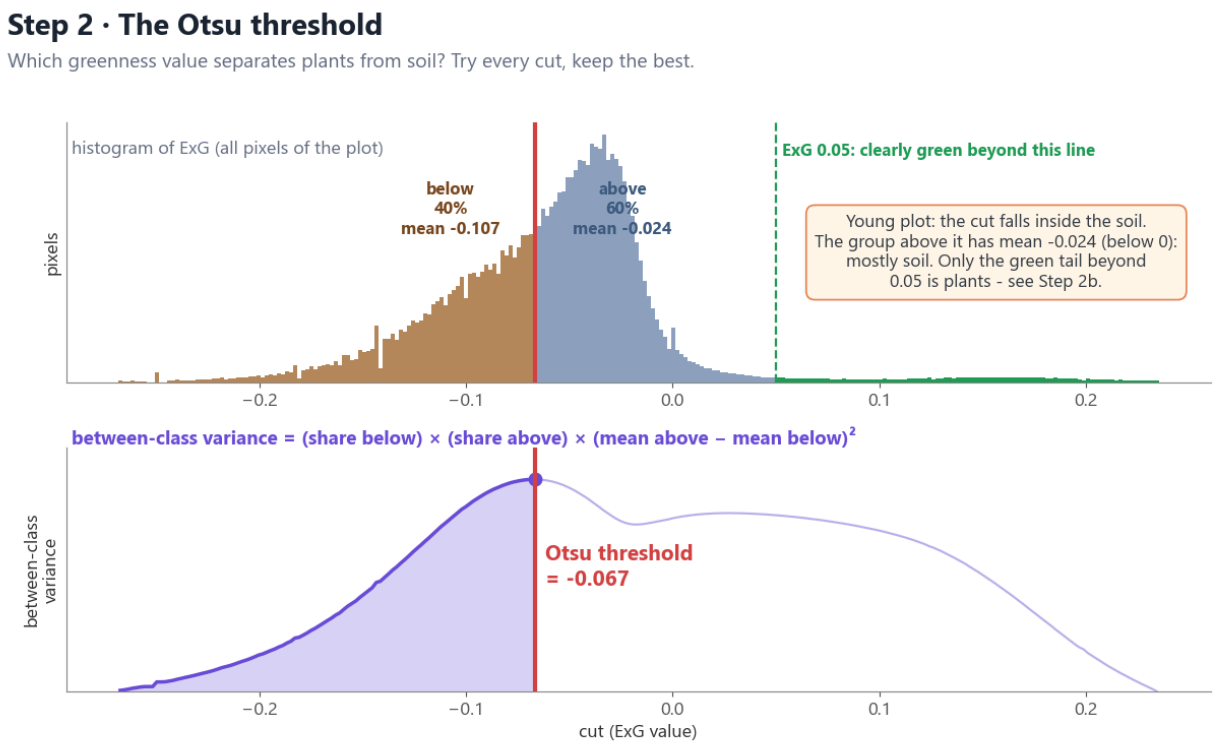

Figure 19. *Otsu's method on the ExG histogram of plot 1001: every cut is tried; the between-class variance (bottom) peaks at the threshold. Brown = below the cut; blue-grey = above the cut but not green (soil); green = clearly green, ExG above 0.05 (dashed line).*

Above which ExG value is a pixel a plant? **Otsu's method** answers this from the histogram alone, with no setting. It tries every possible cut. Each cut splits the pixels into two groups, below and above, and gets a score, the **between-class variance**:

between-class variance = (share below) × (share above) × (mean above − mean below)²

The score is large when both groups are big and their means are far apart. The cut with the largest score is the Otsu threshold: the cut that separates the histogram into the two most distinct groups. It adapts to every plot and every light condition.

What the colours show. Only the bars beyond the dashed line at ExG 0.05 are green: those pixels are clearly plants. On plot 1001 the plants are small, so there is no second hump for them, only a thin tail. Otsu still picks its best cut, at −0.067, inside the soil hump: the group above that cut (blue-grey plus the green tail) has a mean ExG of −0.024, below zero, so it is mostly lighter soil, not plants. Step 2b handles this.

Step 2b · When Otsu splits the soil (green fallback)

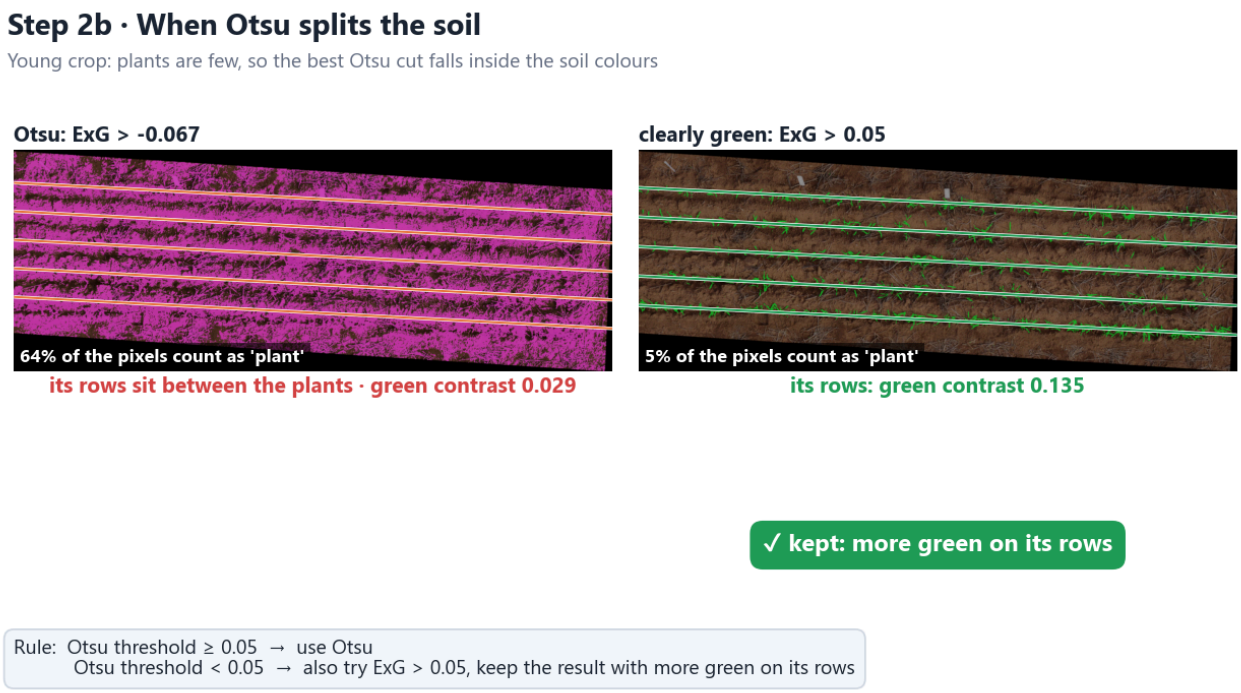

Figure 20. *Left: Otsu's cut (−0.067) falls inside the soil colours; 64% of the pixels count as plant and the rows found sit between the plants. Right: clearly green pixels (ExG > 0.05); its rows sit on the plants and are kept.*

Otsu always finds two groups, even when one of them is not plants. In a young crop the plants are a few percent of the pixels, so the best cut falls inside the soil colours: lighter soil and straw count as plants, and on ridged seedbeds the rows can then be found on the ridges, half a spacing off. So:

- **Otsu threshold ≥ 0.05:** the plants dominate the split; the Otsu result is used.
- **Otsu threshold < 0.05:** the rows are also found from the clearly green pixels only (ExG > 0.05). Both results are scored by how much more green sits on their rows than in the gaps (green contrast), and the higher one is kept. Plot 1001: Otsu 0.029, green 0.135, so the green result is used.

Step 3 · The row profile

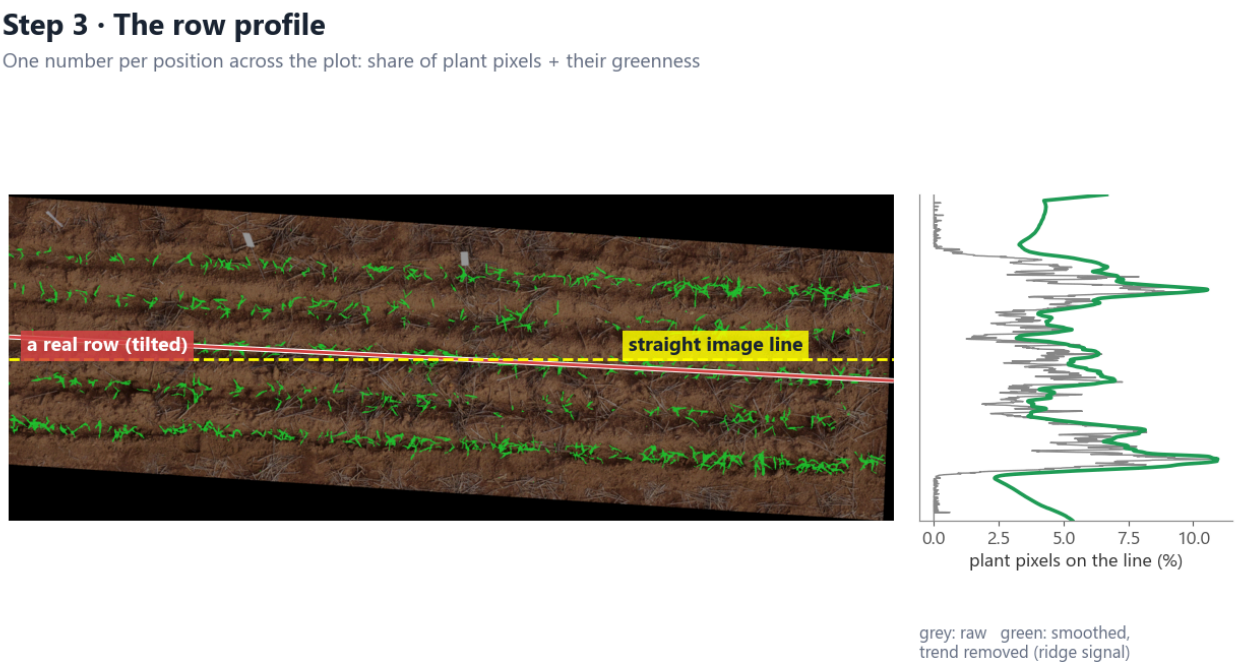

Figure 21. *Right: for each position across the plot, the share of plant pixels along that line (grey) and the smoothed ridge signal (green). The tilted rows smear the peaks of a straight image line.*

For each position across the plot one number is computed along a line in the row direction: the **share of plant pixels** on that line plus their **mean greenness** (the second helps at emergence, when seedlings are only a few pixels wide). Rows become peaks and the soil between them valleys. The profile is smoothed over 4 cm and the slow trend over 0.5 m is removed, so only the row pattern is left (the ridge signal). On a tilted plot a straight image line crosses several rows, so the spacing and the tilt are measured first.

Step 4 · Row spacing from short segments (Fourier)

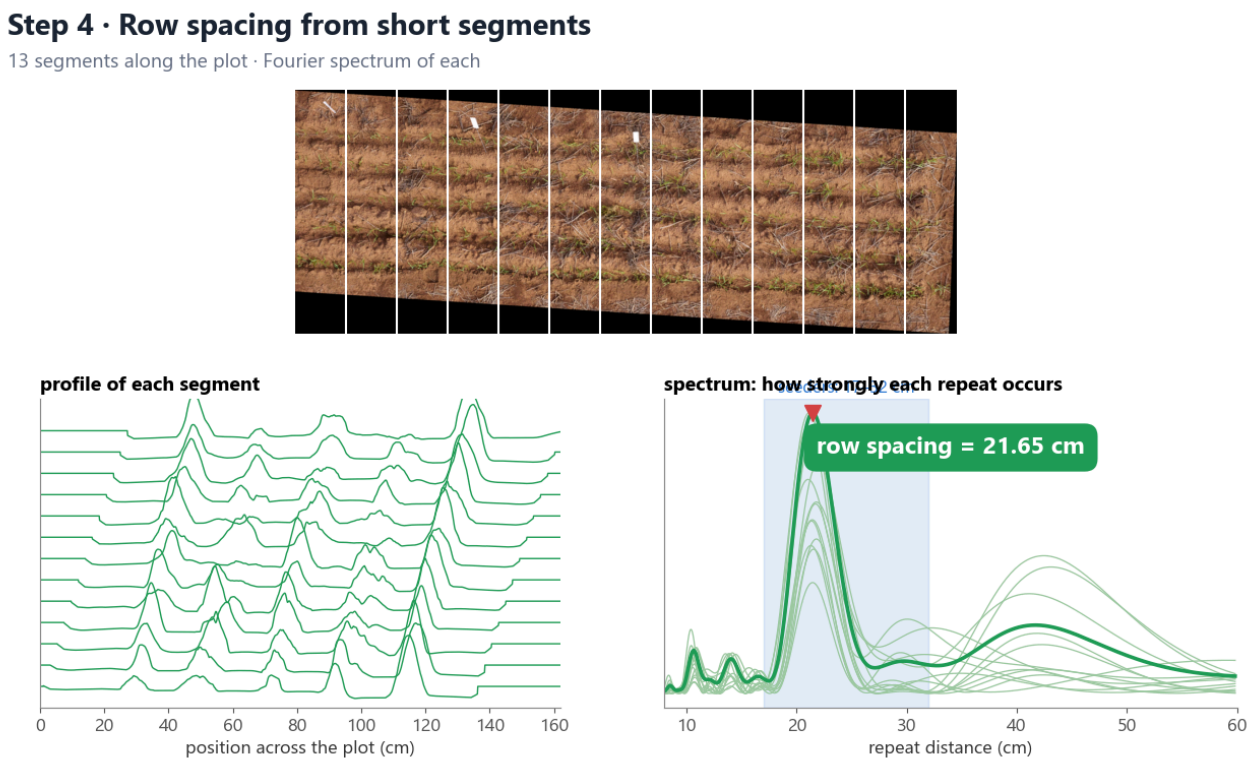

Figure 22. *The plot cut into 13 segments; the profile of each (left) and its spectrum (right). The sum of the spectra peaks at the row spacing.*

The plot is cut into 6 to 24 short segments along its length (13 here). Over a short segment even a tilted row stays sharp. Each segment's profile goes through a **Fourier transform**, which shows how strongly each repeat distance occurs (the position of the rows does not matter there, so the tilt cannot bias it). The spectra are added up and the strongest repeat between **17 and 32 cm** (the plot-seeder range; or exactly the row spacing if you enter one) is the row spacing. It is then refined on the whole, tilt-corrected plot: 21.65 cm here.

Step 5 · Tilt from the drift of the rows

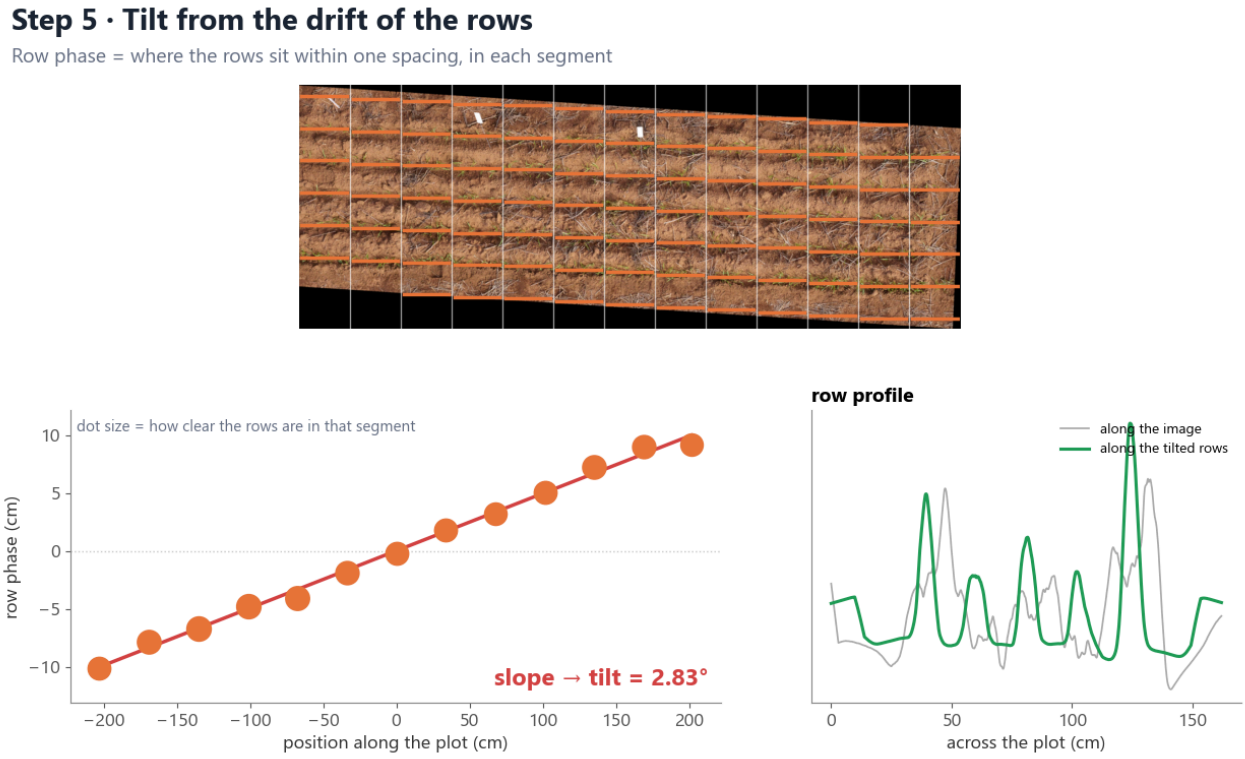

Figure 23. *Top: the row positions measured in each segment. Bottom left: their phase drifts steadily along the plot; the fitted line's slope is the tilt (2.83°). Bottom right: the profile before and after the tilt correction.*

In each segment the **phase** of the rows is measured: where the rows sit within one spacing (from the same Fourier component). If the rows ran exactly along the image, the phase would be the same in every segment; on a tilted plot it drifts steadily along the plot. A straight line is fitted through the phases (weighted by how clear the rows are in each segment, outliers dropped, 4 rounds); its slope is the **tilt** (a tilt above 8° is treated as no tilt). The profile is then measured along the tilted lines, and the row peaks become sharp.

Step 6 · The comb fit

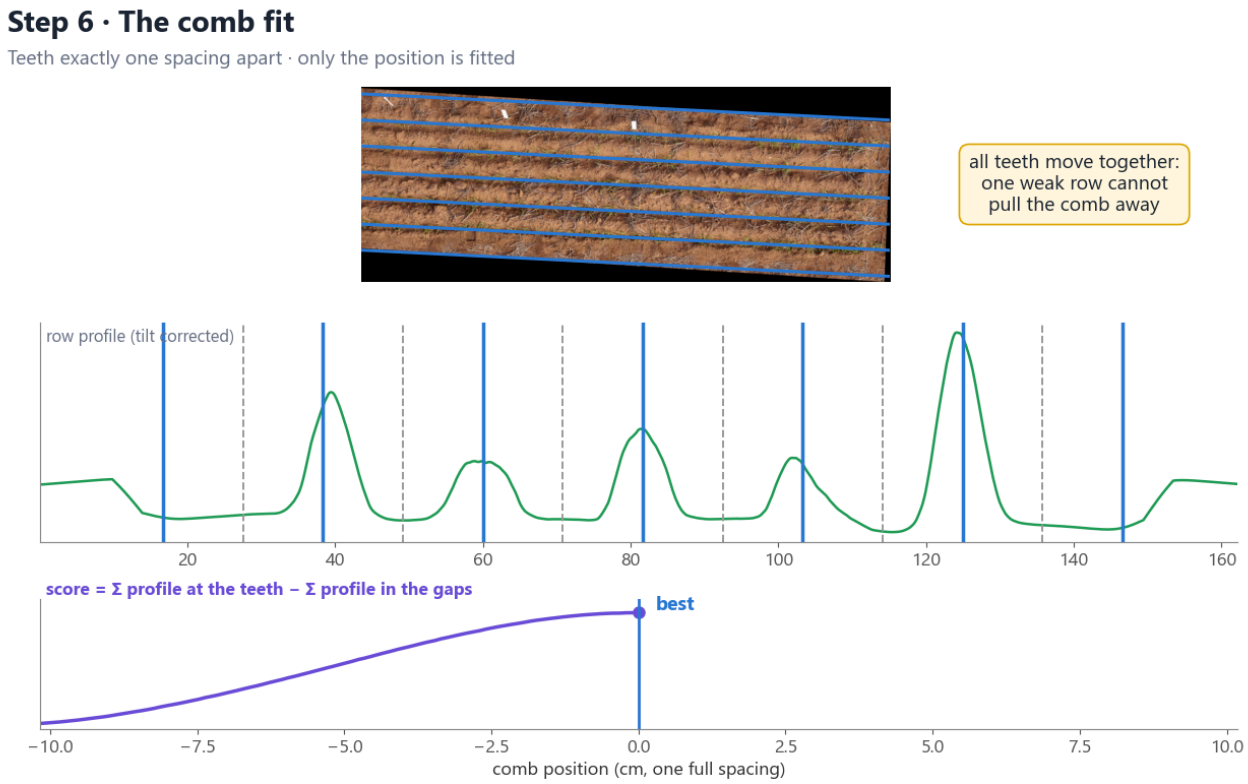

Figure 24. *The comb (blue teeth, one spacing apart) at its best position; dashed lines are the gaps between the teeth. Bottom: the score for every comb position across one spacing.*

A **comb** is a set of teeth exactly one row spacing apart, enough to fill the plot. The spacing is known, so only the comb's position is fitted: it slides across one full spacing, and every position gets a score:

score = Σ profile at the teeth − Σ profile halfway between the teeth

The score is high when the teeth sit on rows and the gaps on soil; the best position wins. Because all teeth move together, one weak, gappy or missing row cannot pull the comb away: the other rows hold it in place. That is why a comb is more reliable than picking the peaks one by one.

Which teeth are the plot's rows

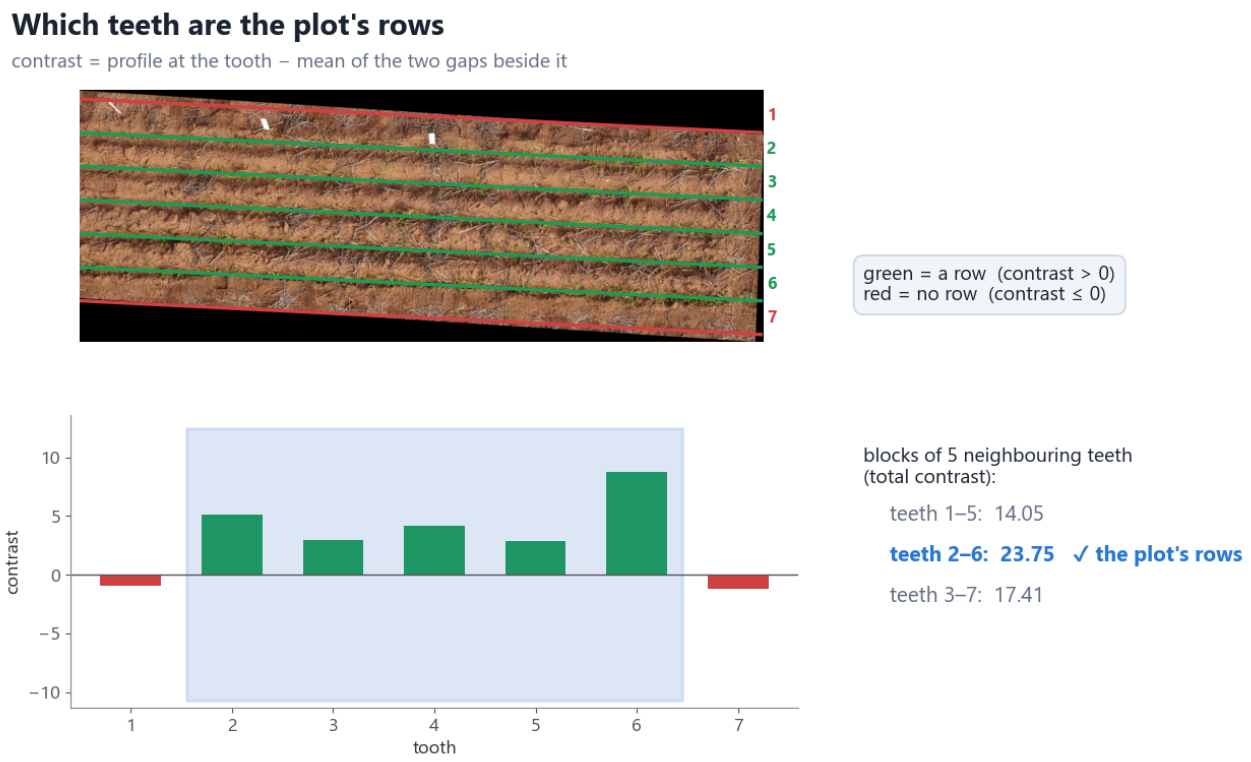

Figure 25. *Contrast of every tooth (green = a row, red = no row) and the block of 5 neighbouring teeth with the highest total contrast.*

Every tooth gets a **contrast**: the profile at the tooth minus the mean of the two gaps beside it. Positive = a real row (green), zero or negative = no row (red, e.g. bare ground at the edge of the crop). With **sown rows per plot** = 5, the block of 5 neighbouring teeth with the highest total contrast is the plot's rows (teeth 2-6 here); this keeps out rows of the neighbouring plots that the margin catches. With sown rows 0 the run of positive teeth is used, without the teeth cut by the crop edge.

Middle rows, band and tiles

Figure 26. *The middle 3 of the 5 rows (yellow), the band (half a spacing beyond the outer two, following the tilt) and the 0.5 m end zones (red). The tiles drawn are band-wide (versions before 2026-10-30); tiles are now 1 m × 1 m centred on the band (Figure in 9.7).*

The **band** of the N middle rows runs from half a row spacing outside the first middle row to half a spacing outside the last, and follows the tilt. The 1 m tiles are centred on it, laid end to end inside the usable length (section 9.6). rows_report.csv lists the rows, spacing and tilt of every plot, and _rows_qa\ has a picture of each.

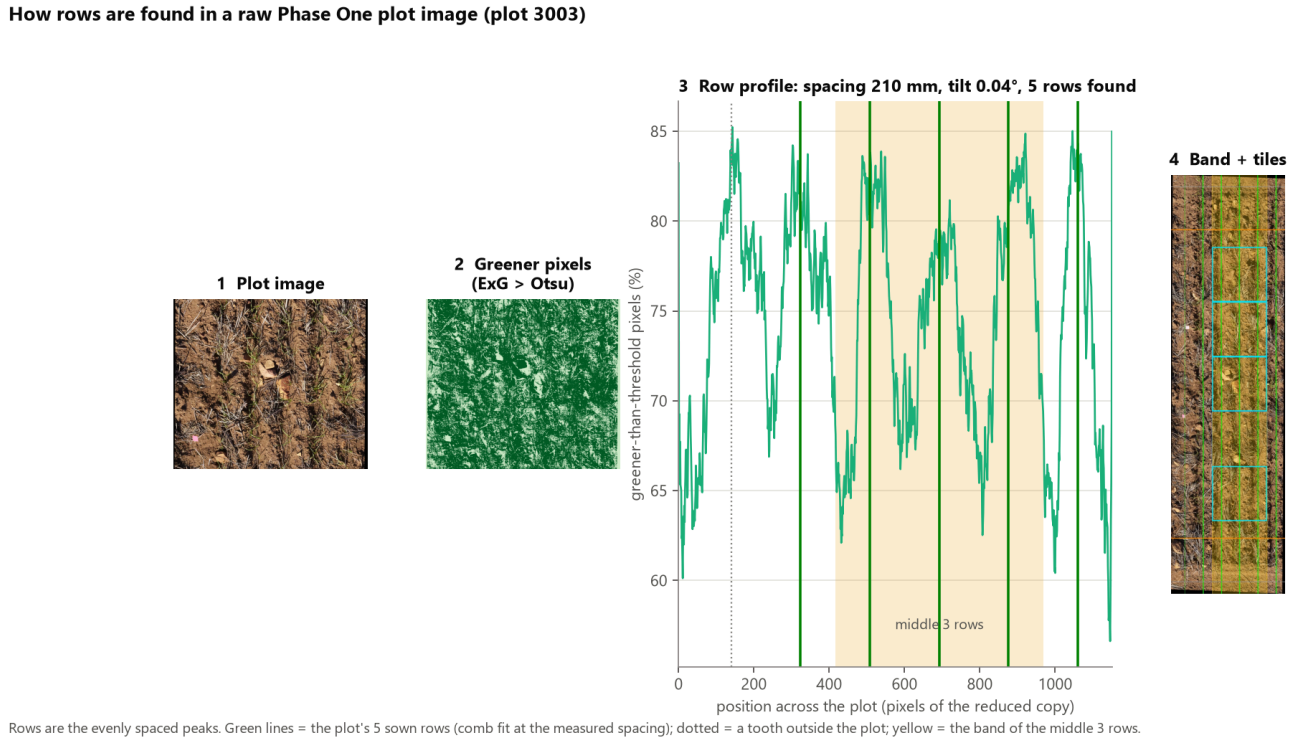

Figure 27. *Summary on another plot (York plot 3003): image, plant pixels, row profile with the fitted rows and the middle band, band and tiles.*

9.6 Where the tiles go along the plot

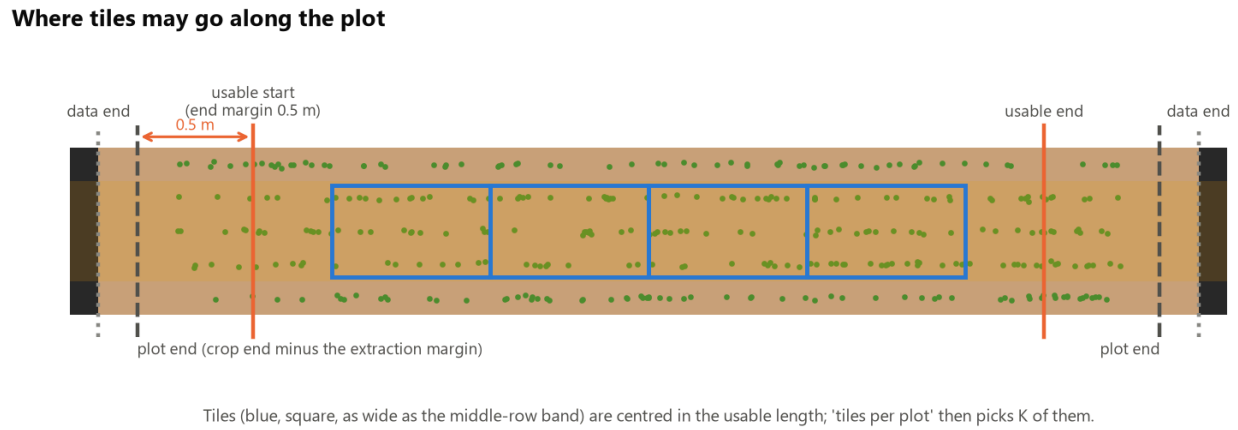

Figure 28. *Tiles stay inside the usable length and are centred in it.*

The plot ends are bare soil (and in a raw crop the black mask or the photo edge). Tiles there would teach a model that empty ground belongs to a plot. So the tiles stay inside the **usable length**: the plot ends (crop ends minus the extraction margin, read from manifest.csv), trimmed to the planted length where the green share clearly drops at the ends, minus the **end distance** you set. The tiles are centred in that length, then **Tiles per plot** picks how many are written.

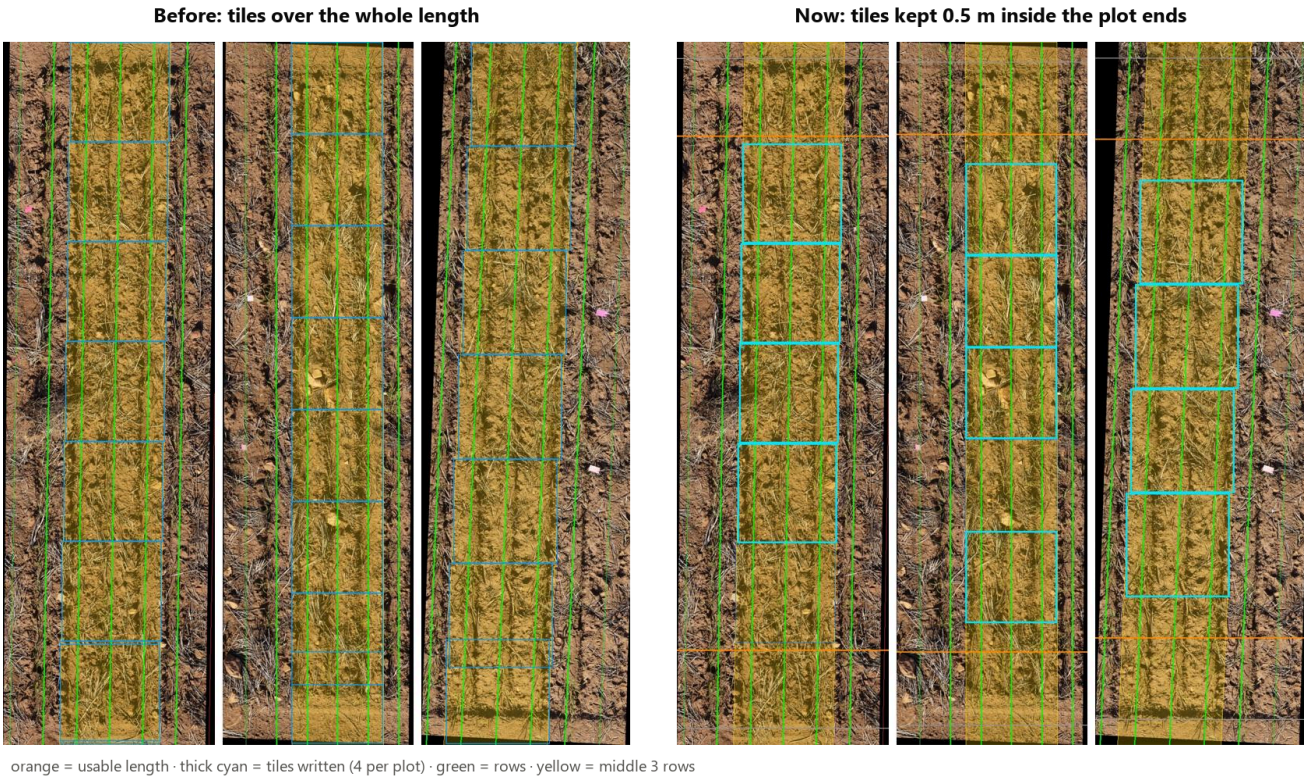

Figure 29. *Before the update tiles ran the whole plot length; now they stay 0.5 m inside the plot ends.*

9.7 What you get (tab 6a)

- <plot>\<plot>_tile_c<i>_r0.tif: the tiles, all the same size (tile size / output scale pixels); exact copies on plots at the output scale, resampled to that size on the others.

- tile_index.csv: per tile, the plot, the pixel window in the crop, the plot and tile mm/px, tile_m and resampled. Needed to map a detection back to its plot.
- rows_report.csv: per plot, rows found, spacing (px and mm), tilt, which rows were used, band, plot length, usable length, tiles that fit, tiles written.
- _rows_qa\<plot>_rows.jpg: green = rows, yellow = middle-row band, grey = plot ends, orange = usable limits, thick cyan = tiles written (numbered), thin blue = tiles not picked.

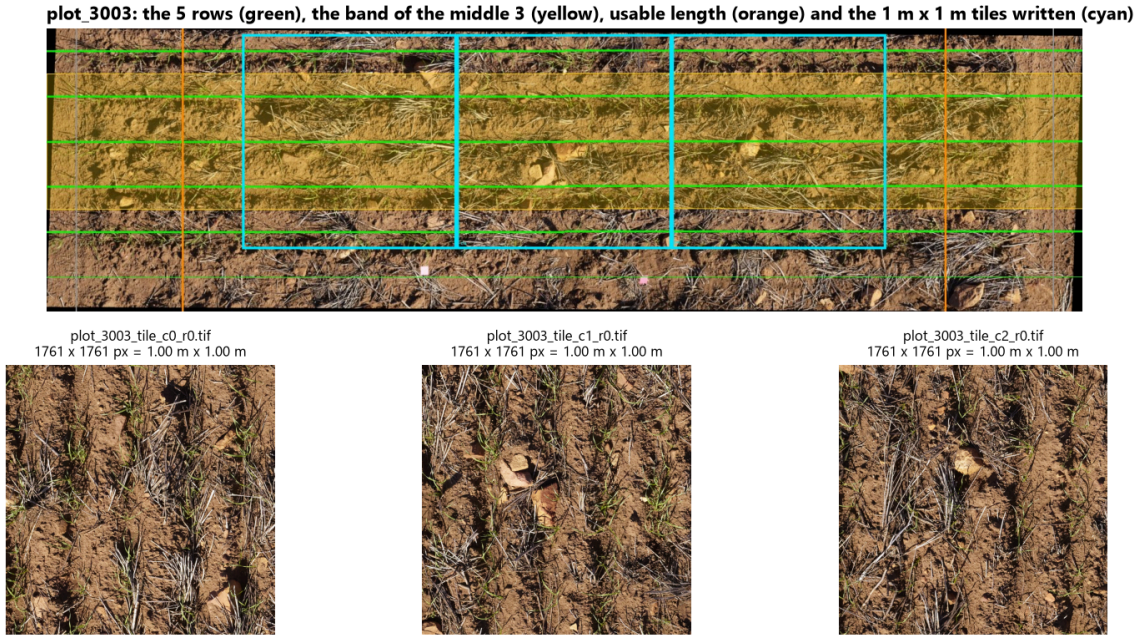

Figure 30. *Plot 3003 (raw): the middle 3 rows and the three 1 m × 1 m tiles that fit between the 0.5 m end zones, laid end to end and centred.*

9.8 Other tools on this tab

- **Plain grid tiles:** untick middle rows. Tiles of the same ground size with overlap over the seed-row band or the whole crop.
- **Rebuild one image per plot:** puts tiles (or labelled overlays with the same names) back together using tile_index.csv.

10. Tab 7 · QA, and are the images really original?

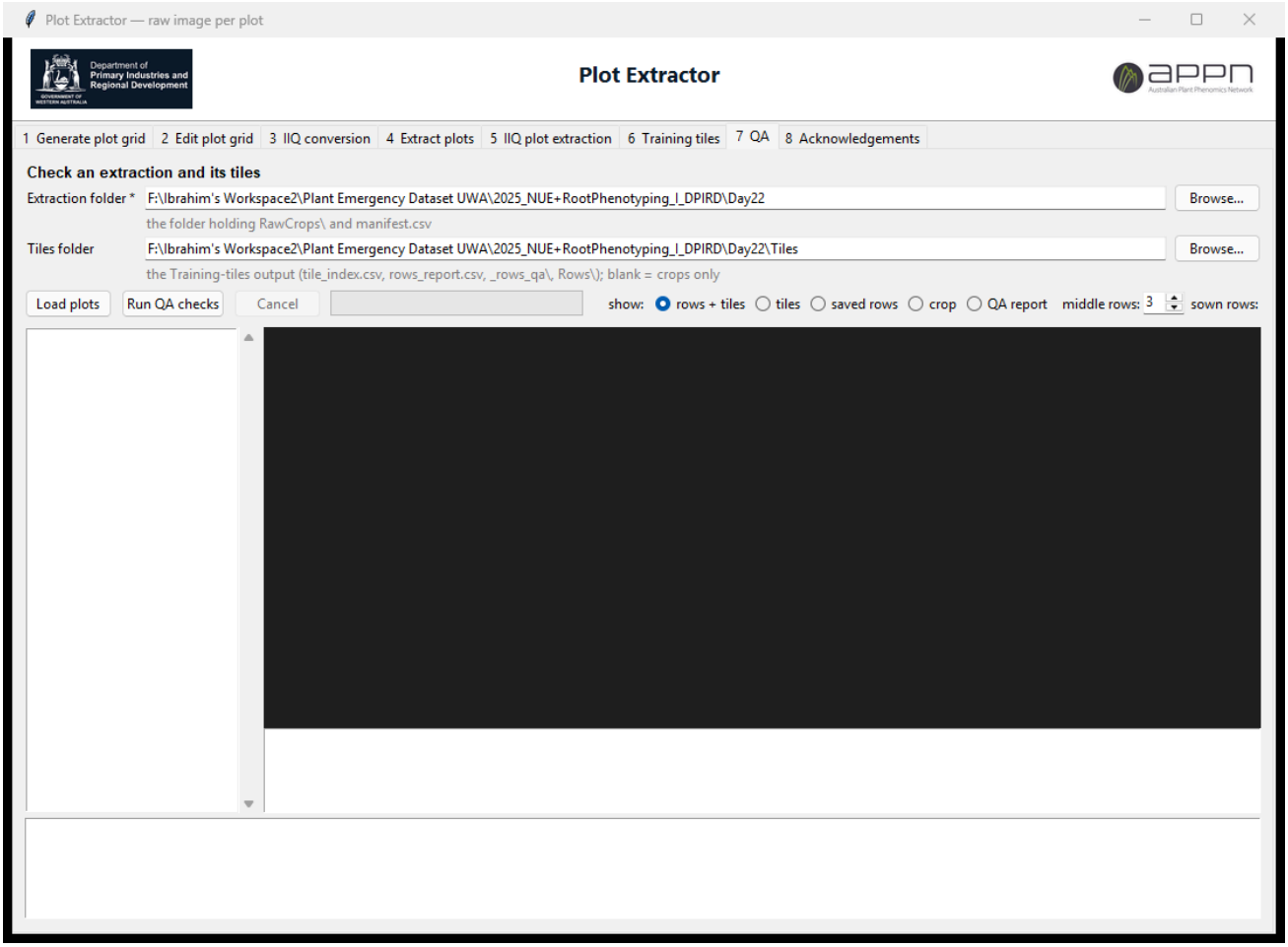

Figure 31. *Tab 7, QA.*

Tab 7 checks an extraction (and its tiles): every raw crop against its source - the JPEG box must be identical; an IIQ crop is compared with a fresh development of its box (IDENTICAL, or MATCH within 2/255 with noise reduction) - every tile against the crop, and the rows. It writes QA\QA_REPORT.txt with CSVs and shows per-plot pictures.

Tested on the York calibration trial (Phase One, 25 June 2025, 36 wheat plots):

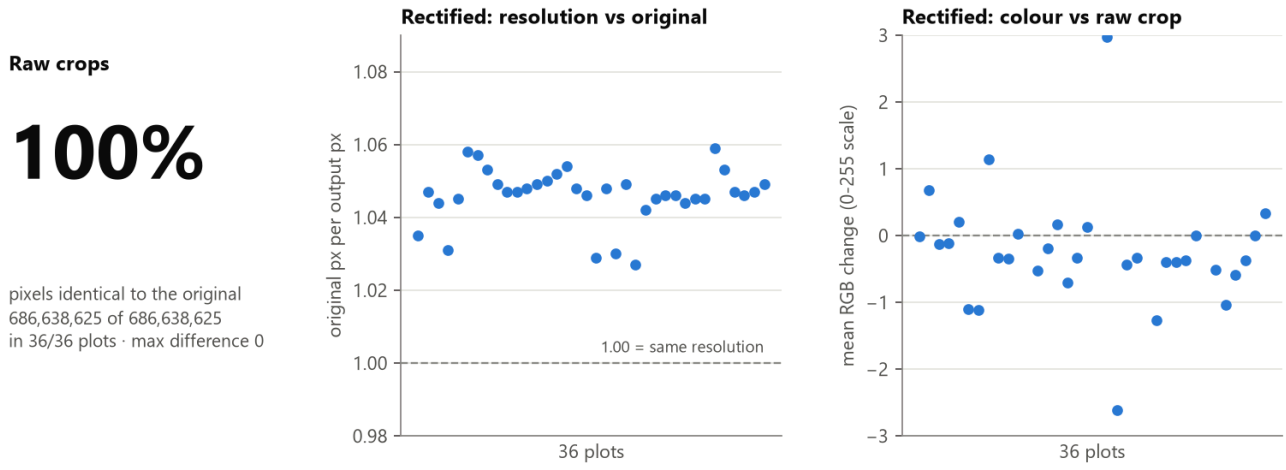

Figure 32. *Raw crops are identical to the original photos; rectified crops are resampled with unchanged colour.*

- **Raw (Masked to plot):** 36/36 plots, 686,638,625 pixels compared with the original JPEGs, every pixel identical. Colour profile copied byte for byte; lossless files.
- **Rectified:** resampled by design, about 1.05 original pixels per output pixel (~5% lower resolution); colour unchanged.
- **Tiles:** every tile is an exact copy of its window in the crop, so raw tiles are original pixels.

The check can be re-run on any extraction with fidelity_check.py (in the York output's fidelity\ folder).

11. Tab 8 · Acknowledgements

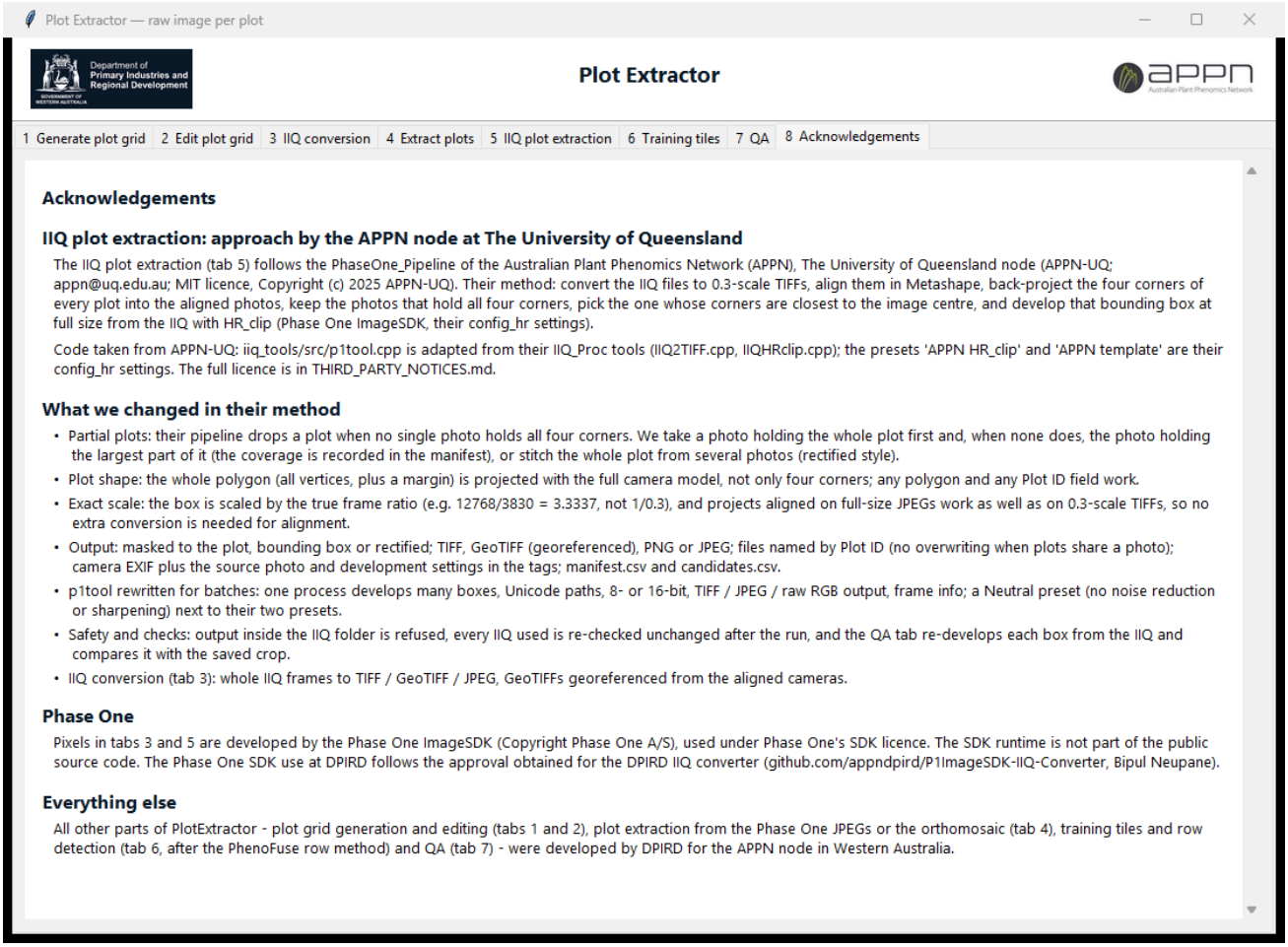

Figure 33. *Tab 8 lists the credits and our changes.*

The IIQ plot extraction (tab 5) follows the **PhaseOne_Pipeline of APPN, The University of Queensland** (appn@uq.edu.au, MIT licence): back-projecting the plot corners into the aligned photos and developing the plot box from the IIQ with HR_clip. Pixels in tabs 3 and 5 are developed by the **Phase One ImageSDK** (Copyright Phase One A/S). Tab 8 lists what we changed in their method: partial plots (largest part) and stitching instead of dropping plots, the whole polygon with a margin, the exact frame scale, masked/rectified GeoTIFF output named by Plot ID, a batch p1tool with a Neutral preset, protection and re-checks of the IIQ files, QA, and whole-frame conversion with georeferencing. Everything else in PlotExtractor was developed by DPIRD for the APPN node in Western Australia.

12. Tips and good practice

12.1 Test first

Run Limit = 3, open the crops and the tile QA pictures, then run everything.

12.2 Use a small margin

A plot grid is rarely exactly on the rows (at York it was about 9 cm off). A 0.10 m margin keeps the outer rows whole, which makes the row choice reliable, and does not add neighbouring rows when the inter-plot gap is ~40 cm.

12.3 Give the sown rows

With the number of sown rows set, all 36 York plots used rows 2-4 of 5 in both the raw and rectified sets. Without it, a row cut by the plot edge can shift which rows count as the middle ones.

12.4 Flight planning for whole plots in one photo

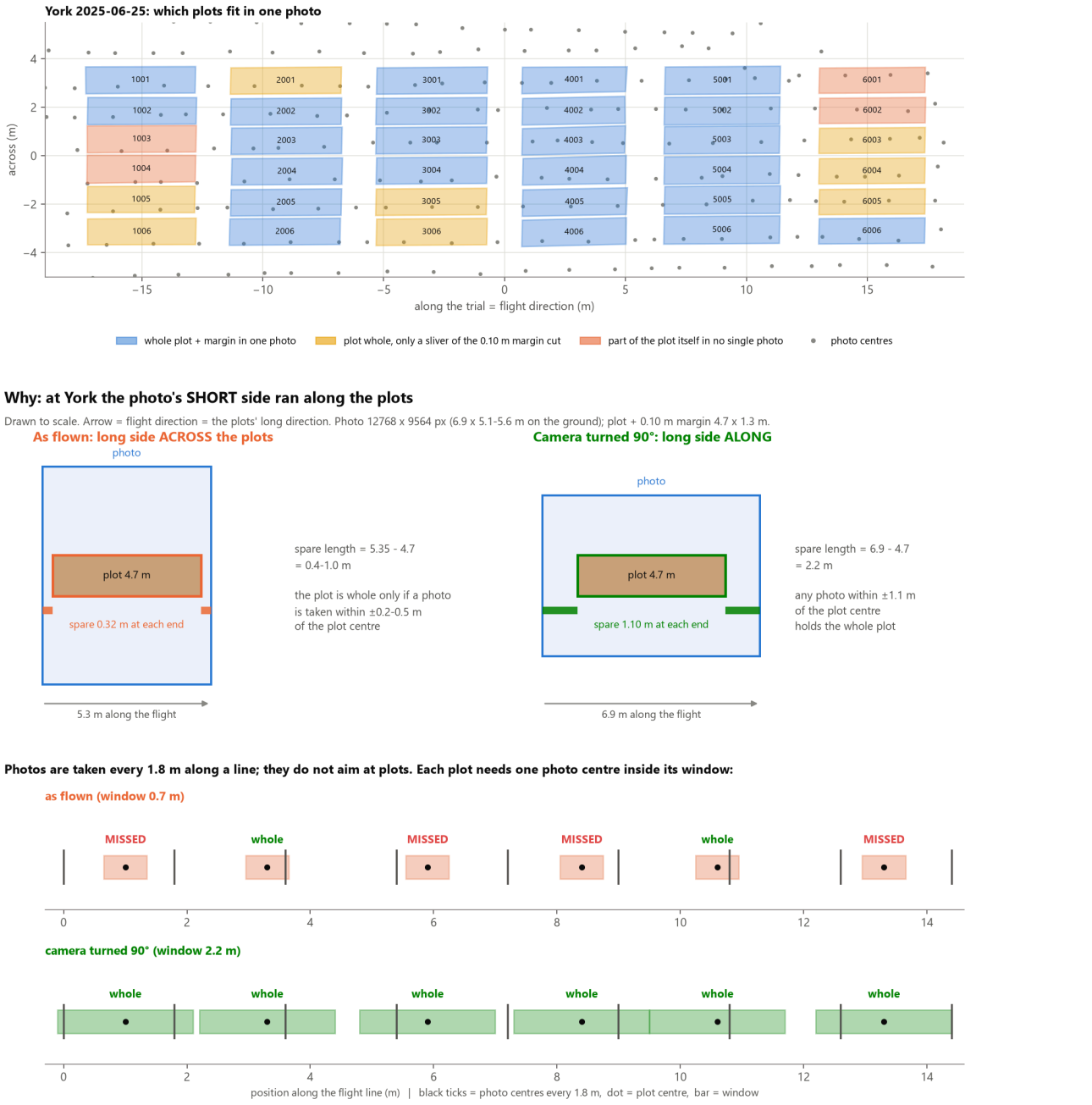

Figure 34. *Top: York plots coloured by fit. Middle: one photo and one plot drawn to scale, as flown and with the camera turned 90°. Bottom: each plot needs a photo centre inside its window; photos come every 1.8 m.*

At York (25 June 2025) the flight lines ran along the plots, but the photo's **short side** (9564 px, 5.1-5.6 m on the ground) lay along them and the long side (12768 px, 6.9 m) across. A plot plus its 0.10 m margin is 4.7 m long, so a photo was only 0.4-1.0 m longer than the plot: the **spare length**. The whole plot is in a photo only if that photo was taken within half the spare length (±0.2-0.5 m) of the plot centre: the plot's **window**.

The camera does not aim at plots: it takes a photo every **1.8 m** along the line (median of 762 photo pairs). A window narrower than 1.8 m catches a photo centre only by luck. Most plots were lucky on at least one flight line; 4 of 36 never were, so their raw images are cut at the photo edge.

Rule: spare length > distance between photos. Spare length = photo length along the plots − plot length (with margin). Then at least one photo centre always falls inside every plot's window, on every flight line.

- **Turn the camera 90°** (rotate it about the vertical axis) so the photo's long side runs along the plots: 6.9 − 4.7 = 2.2 m to spare, more than the 1.8 m between photos, so every plot is whole in at least one photo on every line.
- **Check a flight before processing:** open one photo. The plots must run along its **long** side. (Pointing the camera straight down, −90° gimbal pitch, is a different setting and does not change this.)
- **Or take photos more often:** the distance between photos must be below the spare length (here below 0.4-1.0 m, usually not practical).
- **Or fly 2-3 m higher** (slightly lower resolution).
- **Start and end flight lines a few metres past the trial** so the end plots get the full number of photos.

12.5 Check the QA pictures

Open a handful of _rows_qa pictures: the green lines should sit on the rows, the yellow band on the middle rows and every cyan tile between the orange lines (6a), or every cyan row tile centred on a green line (6b).

12.6 Use one output scale across trials

Within one run every tile has the same pixel size. Runs on different flights have different median scales, so for a training set built from several trials set the same Output scale (mm/px) in every run (e.g. 0.57); the recipe records the value used.

13. Troubleshooting

Message or symptom	Cause and fix
No valid Metashape license	Licence server not reachable: connect to the network/VPN.
plots are ~X km away from the project centre	Wrong CRS or wrong .psx. Check the shapefile's .prj or set Plot CRS.
raw file missing	Set Raw images folder to where the photos are (the JPEG folder).
Chunk 'X' not found	The chunk saved from another project: pick the chunk again (or (active)) after choosing the .psx.
Phase One SDK status not OK (tabs 3, 5)	Leave the SDK folder blank to use the bundled SDK, or point it at an SDK with ImageSdkCpp.dll and SensorProfiles.
No .IIQ file matches the cameras (tab 5)	The IIQ folder must hold the flight's raw files with the same names as the photos in the project (T0_Raw).
frame left without georeference (tab 3)	The camera is not aligned in the chosen chunk, or the project/chunk is not the one the photos were aligned in.
raw_covered_pct below 100	No single photo holds the whole plot. Use the rectified image for that plot, or see 12.4.
no middle-row tiles (only K rows found)	Rows not visible enough or too few rows. Check the QA picture; set sown rows; use a larger extraction margin.
usable length shorter than one tile	Plot too short for the tile size plus end distance: reduce 'keep tiles … m from the plot ends' or the tile size.
smaller than one 1 m tile - skipped / plot is N px across, one tile is M px	The plot image is smaller than one tile in that direction: use a smaller tile size.
rows [k] not in this plot (6b)	A listed row number is beyond the rows found; check sown rows per plot and the QA picture.
no ground scale (6a/6b)	The plot images are not georeferenced: extract as GeoTIFF, or give the plot width (mm).
only K tiles fit, N asked	The plot holds fewer tiles than asked; all that fit are written.
Middle rows look off by one row	Set the sown rows per plot and re-extract with a 0.10 m margin.

Appendix: files

File	Contents
manifest.csv	plot_id, plot_area_m2, margin_m, raw_image, raw_file, crop_w/h, corners_in_frame, raw_covered_pct/m2, cand_frames, best_frame, best_cover_pct/m2, frames_used, seam_y_pct, ortho_*, source_path, raw_style, src_x0, src_y0, src_x1, src_y1, note
candidates.csv	plot_id, rank, camera, cover_pct, cover_m2, corners_in_frame, aniso, mm_per_px, used
tile_index.csv	tile, rel_path, plot_stem, col, row, x0, y0, w, h, out_px, src_mm_px, tile_mm_px, ground_mm, tile_m, resampled, band_lo, band_hi, across_axis, src_w, src_h, veg_pct, src
row_tiles_index.csv (6b)	tile, rel_path, plot_stem, row, of_rows, tile_k, x0, y0, w, h, out_w, out_h, turned_90, resampled, src_mm_px, out_mm_px, tile_len_m, row_width_m, row_centre_px, angle_deg, row_lo_px, row_hi_px, veg_pct, src, bits, src_bits, georef, source photo link
PlotExtractor_recipe.json	The settings of each step (plot_extraction, plot_tiles, rows_and_row_tiles) with the results
rows_report.csv	plot_stem, src, ok, reason, rows_found, rows_present, pitch_px, pitch_mm, angle_deg, middle_rows, band_lo, band_hi, plot_len_m, end_trim, usable_lo, usable_hi, usable_len_m, tiles_fit, tiles_in_band, tiles_drawn